The d-block elements

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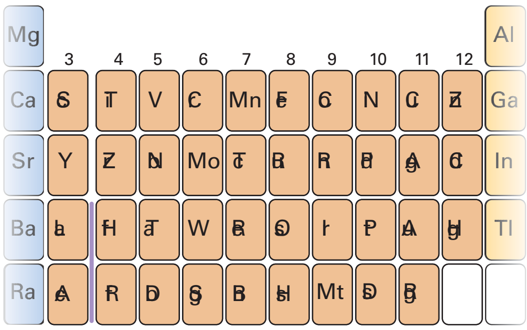


The metallic elements are the most numerous of the elements and of the metallic elements the d-block elements are the most important: their chemical properties are central to both industry and contemporary research. We shall see systematic trends across the block and trends in their chemical properties within each group. One of the most important trends is the variation in stability of the oxidation states because these stabilities are closely related to the ease of recovery of metals from their ores and to the ways in which the various metals and their compounds are handled in the laboratory. It will be seen that simple binary compounds, such as metal halides and oxides, often follow systematic trends; however, when the metal is in a low oxidation state, interesting variations such as metal–metal bonding may be encountered.

The elements
19.1 Occurrence and recovery
19.2 Physical properties

Trends in chemical properties
19.3 Oxidation states across a series
19.4 Oxidation states down a group
19.5 Structural trends
19.6 Noble character

Representative compounds
19.7 Metal halides
19.8 Metal oxides and oxido complexes
19.9 Metal sulfides and sulfide complexes
19.10 Nitrido and alkylidyne complexes
19.11 Metal–metal bonded compounds and clusters

FURTHER READING
EXERCISES
PROBLEMS

The two terms **d-block metal** and **transition metal** are often used interchangeably; however, they do not mean the same thing. The name transition metal originally derived from the fact that their chemical properties were transitional between those of the s and p blocks. Now, however, the IUPAC definition of a **transition element** is that it is an element that has an incomplete d subshell in either the neutral atom or its ions. Thus the Group 12 elements (Zn, Cd, Hg) are members of the d block but are not transition elements. In the following discussion, it will be convenient to refer to each row of the d block as a series, with the 3d series the first row of the block (Period 4), the 4d series the second row (Period 5), and so on. It will prove important to note the intrusion of the f block, the **inner transition elements**, the lanthanoids, into the 5d series. Elements towards the left of the d block are often referred to as *early* and those towards the right are referred to as *late*.

The elements

The chemical properties of the d-block elements (or d metals as we shall commonly call them) will occupy this and the next three chapters. It is therefore appropriate to begin with a survey of their occurrence and properties. The trends in their properties and the correlation with their electronic structures and therefore location in the periodic table should be kept in mind throughout these chapters.

19.1 Occurrence and recovery

Key point: Chemically soft members of the block occur as sulfide minerals and are partially oxidized to obtain the metal; the more electropositive 'hard' metals occur as oxides and are extracted by reduction.

The elements on the left of the 3d series occur in nature primarily as metal oxides or as metal cations in combination with oxoanions (Table 19.1). Of these elements, titanium ores are the most difficult to reduce, and the element is widely produced by heating TiO_2 with chlorine and carbon to produce $TiCl_4$, which is then reduced by molten magnesium at about 1000°C in an inert-gas atmosphere. The oxides of Cr, Mn, and Fe are reduced with carbon (Section 5.16), a much cheaper reagent. To the right of Fe in the 3d series, Co, Ni, Cu, and Zn occur mainly as sulfides and arsenides, which is consistent with the increasingly soft Lewis acid character of their dipositive ions. Sulfide ores are usually roasted in air either to the metal directly (for example, Ni) or to an oxide that is subsequently reduced (for example, Zn). Copper is used in large quantities for electrical conductors; electrolysis is used to refine crude copper to achieve the high purity needed for high electrical conductivity.

The difficulty of reducing the early 4d and 5d metals Mo and W is apparent from Table 19.1. It reflects the tendency of these elements to have stable high oxidation states, as discussed later. The platinum metals (Ru and Os, Rh and Ir, and Pd and Pt), which are found at the lower right of the d block, occur as sulfide and arsenide ores, usually in association with larger quantities of Cu, Ni, and Co. They are collected from the sludge that forms during the electrolytic refinement of copper and nickel. Gold (and to some extent silver) is found in its elemental form.

The role that certain d metals play in the environment is summarized in Box 19.1.

19.2 Physical properties

Key points: The properties of the d metals are largely derived from their electronic structure, with the strength of metallic bonding peaking at Group 6; the lanthanide contraction is responsible for some of the anomalous behaviour of the metals in the 5d series.

The d block of the periodic table contains the metals most important to modern society. It contains the immensely strong and light titanium, the major components of most steels (Fe, Cr, Mn, Mo), the highly electrically conducting copper, the malleable gold and platinum, and the very dense osmium and iridium. To a large extent these properties derive from the nature of the metallic bonding that binds the atoms together.

Table 19.1 Mineral sources and methods of recovery of some commercially important d metals

Metal	Principal minerals	Method of recovery	Note
Titanium	Ilmenite, $FeTiO_3$ Rutile, TiO_2	$TiO_2 + 2C + 2Cl_2 \rightarrow TiCl_4 + 2CO$ followed by reduction of $TiCl_4$ with Na or Mg	
Chromium	Chromite, $FeCr_2O_4$	$FeCr_2O_4 + 4C \rightarrow Fe + 2Cr + 4CO$	(a)
Molybdenum	Molybdenite, MoS_2	$2MoS_2 + 7O_2 \rightarrow 2MoO_3 + 4SO_2$ followed by either $MoO_3 + 2Fe \rightarrow Mo + Fe_2O_3$ or $MoO_3 + 3H_2 \rightarrow Mo + 3H_2O$	
Tungsten	Scheelite, $CaWO_4$ Wolframite, $FeMn(WO_4)_2$	$CaWO_4 + 2HCl \rightarrow WO_3 + CaCl_2 + H_2O$ followed by $WO_3 + 3H_2 \rightarrow W + 3H_2O$	
Manganese	Pyrolusite, MnO_2	$MnO_2 + 2C \rightarrow Mn + 2CO$	(b)
Iron	Haematite, Fe_2O_3 Magnetite, Fe_3O_4 Limonite, $FeO(OH)$	$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$	
Cobalt	CoAsS Smaltite, $CoAs_2$ Linnaeite, Co_3S_4	Byproduct of copper and nickel production	
Nickel	Pentlandite, $(Fe,Ni)_6S_8$	$NiS + O_2 \rightarrow Ni + SO_2$	(c)
Copper	Chalcopyrite, $CuFeS_2$ Chalocite, Cu_2S	$2CuFeS_2 + 2SiO_2 + 5O_2 \rightarrow 2Cu + 2FeSiO_3 + 4SO_2$	

(a) The iron–chromium alloy is used directly for stainless steel.
(b) The reaction is carried out in a blast furnace with Fe_2O_3 to produce alloys.
(c) NiS is formed by smelting the ore and is then separated by physical processes. NiO is used in a blast furnace with iron oxides to produce steel. Nickel is purified by electrolysis or the Mond process via $Ni(CO)_4$, Box 22.1.

BOX 19.1 Toxic metals in the environment

The biosphere evolved in close association with all the elements in the periodic table, and has had a long time to adapt to them. Metals are released from rocks by weathering, and are processed by a variety of mechanisms, including biological ones. Indeed, many metals have been harnessed for essential biochemical functions (Chapter 27), and other biochemical systems have evolved to sequester metals and keep them out of harm's way. However, the natural biogeochemical cycles have been greatly perturbed by mining: circulating levels of most metals have increased dramatically since pre-industrial times.

Many metals and metalloids, including Be, Mn, Cr, Ni, Cd, Hg, Pb, Se, and As, are hazardous in occupational or environmental settings. Exposure to these elements varies widely, and depends on the pattern of their industrial use, and on their environmental chemistry. The metals of greatest environmental concern are heavy metals, such as mercury, which, being chemically very soft, bind tightly to thiol groups in proteins.

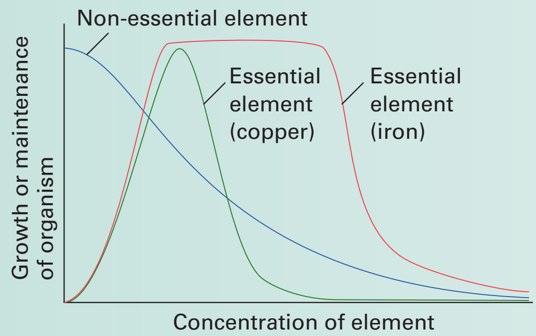


Figure B19.1 The idealized dose−response curves for a non-essential element and two essential elements.

The effect of a substance on a living organism is often plotted as a *dose−response curve*, such as those in Fig. B19.1. The shape of the dose−response curves for various metals varies widely, depending on physiological variables, and on the chemistry of the metals. For example, iron and copper are both essential elements but have very different dose−response curves: toxicity occurs at a much lower concentration for copper than for iron. It is reasonable to associate this toxicity with the higher affinity of copper ions for sulfur and nitrogen ligands, which makes copper more likely to interfere with crucial sites in proteins. Nevertheless, iron is harmful at higher doses because it can catalyse the production of oxygen radicals, and also because excess iron can stimulate the growth of bacteria.

The ability of elements to exist in alternative redox states can be of crucial importance to the hazard they present. For example, the solubilities of different oxidation states of a particular metal are often very different. Many metals are found as insoluble sulfides in the Earth's crust and are thus immobile. However, when they are disturbed and brought into contact with air, which oxidizes them, they can become very mobile. An example is mercury(II) sulfide, which is oxidized to the mobile mercury(II) sulfate on exposure to air.

Another effect of oxidation is exemplified by iron, which occurs widely as the mineral pyrite, FeS_2. Mining operations often lead to acid drainage due to the reaction

$$FeS_2 + \tfrac{15}{4}O_2 + \tfrac{7}{2}H_2O \rightarrow Fe(OH)_3 + 2SO_4^{2-} + 4H^+$$

In this process (which is generally catalysed by certain bacteria), S_2^{2-} is oxidized to sulfuric acid and Fe^{2+} is oxidized to Fe^{3+}, which hydrolyses to $Fe(OH)_3$, yielding additional H^+ ions. Streams emanating from iron mines, many of them abandoned, are thus often highly acidic.

Redox status can have opposite effects on solubility for different metals. Thus, whereas manganese is mobile under reducing conditions, other metals are mobilized under oxidizing conditions. For instance, Cr(III) is immobile, forming an insoluble oxide or kinetically inert complexes with soil constituents; in addition, there is no mechanism for transporting free Cr(III) ions across biological membranes. However, Cr(III) is solubilized upon oxidation to the chromate(VI) ion, CrO_4^{2-}. Chromate(VI) is highly toxic, and is implicated in cancer. The CrO_4^{2-} ion has the same shape and charge as SO_4^{2-}, and can be ferried across biological membranes by sulfate transport proteins. Once inside a cell, chromate is converted to Cr(III) complexes by endogenous reductants, such as ascorbate. In the process, reactive oxygen species are generated, which can damage DNA; in addition, the Cr(III) reduction product can complex and cross-link DNA. Cr(III) accumulates inside the chromate-exposed cells, as it does not have a mechanism for transport through biological membranes.

For other metals, toxicity is controlled by the interplay of membrane transport and intracellular binding with the redox state. For example, mercury salts are relatively harmless because membranes, which present a barrier to ionic species, are impermeable to Hg^{2+}. Likewise, metallic mercury is not absorbed through the gut, so it is not toxic when swallowed. However, mercury vapour is highly toxic (that is why all mercury spills must be rigorously cleaned up) because the neutral atoms readily pass through the lungs' membranes and also across the blood−brain barrier. Once in the brain, the Hg(0) is oxidized to Hg(II) by the vigorous oxidizing activity of brain cell mitochondria, and the Hg(II) binds tightly to crucial thiolate groups of neuronal proteins. Mercury is a powerful neurotoxin, but only if it gets inside nerve cells. Even more hazardous than mercury vapour are organomercury compounds, particularly methylmercury. Thus, CH_3Hg^+ is taken up through the gut because it is complexed by the stomach's chloride, forming CH_3HgCl, which, being electrically neutral, can pass through a membrane. Once inside cells, CH_3Hg^+ binds to thiolate groups and accumulates.

The environmental toxicity of mercury is associated almost entirely with eating fish. Methylmercury is produced by the action of sulfate-reducing bacteria on Hg^{2+} in sediments, and accumulates as little fish are eaten by bigger fish further up the aquatic food chain. Fish everywhere have some level of mercury present. Mercury levels can increase markedly if sediments are contaminated by additional mercury. The worst known case of environmental mercury poisoning occurred in the 1950s in the Japanese fishing village of Minamata. A polyvinyl chloride plant, using Hg^{2+} as a catalyst, discharged mercury-laden residues into the bay, where fish accumulated methylmercury to levels approaching 100 ppm. Thousands of people were poisoned by eating the fish, and a number of infants suffered mental disabilities and motor disturbance from exposure *in utero*. This disaster led to strict standards for fish consumption. Limited consumption is advised for fish at the top of the food chain, such as pike and bass in fresh waters, and swordfish and tuna in the oceans.

Regulatory action has been aimed at reducing mercury discharges and emissions, and industrial point sources have been largely controlled. Chloralkali plants, producing the large-volume industrial chemicals Cl_2 and NaOH by electrolysis of NaCl, were a major source as a mercury pool electrode was used to transfer metallic sodium to a separate hydroxide-generating compartment. However, this is now accomplished by separating the two electrode compartments with a cation-exchange membrane, which prevents migration of the anions. Combustion can vent mercury to the atmosphere if the fuel contains mercury compounds. Whereas municipal waste and hospital incinerators have been equipped with filters to reduce

(*Continued*)

BOX 19.1 *(Continued)*

mercury emissions to the air, coal contains small amounts of mercury minerals. Because of the huge quantities burned, coal is a major contributor to environmental mercury.

Mercury is a global problem because mercury vapour and volatile organomercurial compounds can travel long distances in the atmosphere. Eventually elemental mercury is oxidized and organomercurials are decomposed, both to Hg^{2+}, by reaction with ozone or by atmospheric hydroxyl or halogen radicals. The Hg^{2+} ions are solvated by water molecules and deposited in rainfall. Thus mercury deposition can occur far from the emission source, and is distributed fairly uniformly around the globe. For example, it is estimated that only a third of North American mercury emissions are in fact deposited in the USA, and this accounts for only half of the mercury deposition in the USA. Even the gold fields of Brazil contribute because miners use mercury to extract the gold, which is recovered by heating the resultant amalgam to drive off the mercury. This practice is estimated to account for 2 per cent of global mercury emissions (half of South American emissions). The picture is further complicated by the fact that much of the deposited mercury is recirculated through processes that produce volatile compounds or mercury vapour. For example, much of the biomethylation activity of the sulfate-reducing bacteria produces dimethylmercury, $(CH_3)_2Hg$, which, being volatile, is vented to the atmosphere. Other bacteria have an enzyme (methylmercury lyase) that breaks the methyl—mercury bond of CH_3Hg^+, and another enzyme (methylmercury reductase) that reduces the resulting Hg(II) to Hg(0); this is a protective mechanism for the microorganisms, ridding them of mercury as volatile Hg(0).

Metallic bonding was covered in Chapter 3, which introduced the concept of band structure. Generally speaking, the same band structure is present for all the d-block metals and arises from the overlap of the $(n+1)$s orbitals to give an s band and of the nd orbitals to give a d band. The principal differences between the metals is the number of electrons available to occupy these bands: Ti ($3d^2 4s^2$) has four bonding electrons, V ($3d^3 4s^2$) five, Cr ($3d^5 4s^1$) six, and so on. The lower, net bonding region of the valence band is therefore progressively filled with electrons on going to the right across the block, which results in stronger bonding, until around Group 7 (at Mn, Tc, Re) when the electrons begin to populate the upper, net antibonding part of the band. This trend in bonding strength is reflected in the increase in melting point from the low-melting alkali metals (effectively only one bonding electron for each atom, resulting in melting points typically less than 100°C) up to Cr, and its decline thereafter to the low-melting Group 12 metals (mercury being a liquid at room temperature, Fig. 19.1 and Section 9.2). The strength of metallic bonding in tungsten is such that its melting point (3410°C) is exceeded by only one other element, carbon.

The radii of d-metal ions depend on the effective charge of the nucleus, and ionic radii generally decrease on moving to the right as the atomic number increases. The radius of the metal atoms in the solid element is determined by a combination of the strength of the metallic bonding and the size of the ions. Thus, the separations of the centres of the atoms in the solid generally follow a similar pattern to the melting points: they decrease to the middle of the d block, followed by an increase back up to Group 12, with the smallest separations occurring in and near Groups 7 and 8.

The atomic radii of the elements in the 5d series (Hf, Ta, W,…) are not much bigger than those of their 4d-series congeners (Zr, Nb, Mo,…). In fact, the atomic radius of Hf

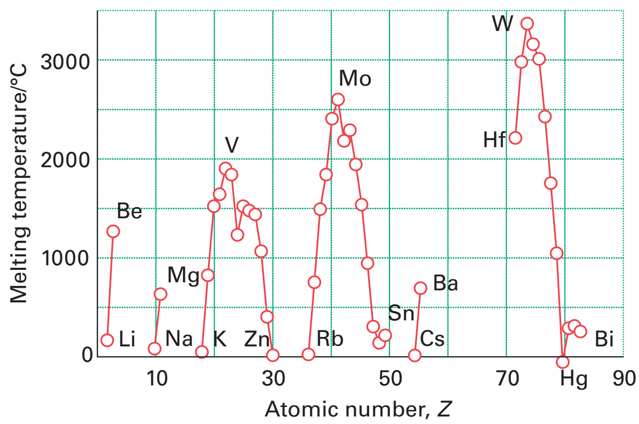


Figure 19.1 The melting points of the metals in Groups 1—15.

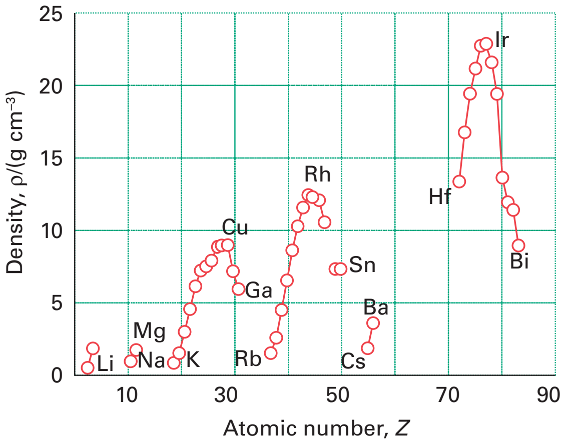


Figure 19.2 The densities of the elements in Groups 1–15.

is smaller than that of Zr even though it appears in a later period. To understand this anomaly, we need to consider the effect of the lanthanoids (the first row of the f block).

The intervention of the lanthanoid elements in Period 6 corresponds to the occupation of the poorly shielding 4f orbitals. Because the atomic number has increased by 32 between Zr in Period 5 and its congener Hf in Period 6 without a corresponding increase in shielding, the overall effect is that the atomic radii of the 5d-series elements are much smaller than expected. This reduction in radius is the *lanthanide contraction* introduced in Section 1.9a. The lanthanide contraction also affects the ionization energies of the 5d-series elements, making them higher than expected on the basis of a straightforward extrapolation. Some of the metals—specifically Au, Pt, Ir, and Os—have such high ionization energies that they are unreactive under normal conditions.

Atomic mass increases with atomic number, and the combination of this increase with the changes in the radii of the metal atoms in the metal lattice means that the mass densities of the elements reach a peak with Ir (density 22.65 g cm^{-3}). Figure 19.2 illustrates this trend.

Trends in chemical properties

Many of the d metals display a wide range of oxidation states, which leads to a rich and fascinating chemistry. They also form an extensive range of coordination compounds (Chapters 7, 20, and 21) and organometallic compounds (Chapter 22). Many of the trends discussed in Chapter 9 are applicable to the d metals; in particular we should recall that ionization energies generally decrease down a group and increase across a period.

19.3 Oxidation states across a series

The range of oxidation states of the d metals accounts for the interesting electronic properties of many solid compounds (Chapters 24 and 25), their ability to participate in catalysis (Chapter 26), and their subtle and interesting role in biochemical processes (Chapter 27). In this section we focus on trends in the stabilities of oxidation states within the d block.

(a) High oxidation states

Key point: The group oxidation state can be achieved by elements that lie towards the left of the d block but not by elements on the right. Oxygen is usually more effective than fluorine at bringing out the highest oxidation states because less crowding is involved.

The group oxidation state (in which the oxidation number is equal to the Group number) can be achieved by elements that lie towards the left of the d block but not by elements on the right (Section 9.5). For example, Sc, Y, and La in Group 3, with configurations $nd^1(n+1)s^2$, are found in aqueous solution only in oxidation state +3, which corresponds to the loss of all their outermost electrons, and the majority of their complexes contain the elements in this state. The group oxidation state is never achieved after Group 8 (Fe, Ru, and Os). This limit on the maximum oxidation state correlates with the increase in ionization energy and hence noble character from left to right across each series in the d block.

The trend in thermodynamic stability of the group oxidation states of the 3d-series elements is illustrated in Fig. 19.3, which shows the Frost diagram for species in aqueous acidic solution. We see that the group oxidation states of Sc, Ti, and V fall in the lower part of the diagram. This location indicates that the element and any species in intermediate oxidation states are readily oxidized to the group oxidation state. By contrast, species in the group oxidation state for Cr and Mn (+6 and +7, respectively) lie in the upper part of the diagram. This location indicates that they are very susceptible to reduction. The Frost diagram shows that the group oxidation state is not achieved in Groups 8−12 of the 3d series (Fe, Co, Ni, Cu, and Zn), and also shows the oxidation states that are most stable under acid conditions; namely Ti^{3+}, V^{3+}, Cr^{3+}, Mn^{2+}, Fe^{2+}, Co^{2+}, and Ni^{2+}.

The binary compounds of the 3d-series elements with the halogens and with oxygen also illustrate the trend in stabilities of the group oxidation states. The earliest metals can achieve their group oxidation states in compounds with chlorine (for example, $ScCl_3$ and $TiCl_4$), but the more strongly oxidizing halogen fluorine is necessary to achieve the group oxidation state of V (Group 5) and Cr (Group 6), which form VF_5 and CrF_6, respectively. Beyond Group 6 in the 3d series, even fluorine cannot produce the group oxidation state, and MnF_7 and FeF_8 have not been prepared. Oxygen brings out the group oxidation state for many elements more readily than does fluorine because fewer O atoms than F atoms are needed to achieve the same oxidation number, thus decreasing steric crowding. For example, the group oxidation state of +7 for Mn is achieved in manganate(VII) salts, such as potassium permanganate, $KMnO_4$.

As can be inferred from the Frost diagram in Fig. 19.3, chromate(VI) CrO_4^{2-}, manganate(VII) MnO_4^-, and ferrate(VI) FeO_4^{2-} are strong oxidizing agents and become stronger from CrO_4^{2-} to FeO_4^{2-}. This trend is another illustration of the decreasing stability of the maximum attainable oxidation state for Groups 6, 7, and 8. Yet another example of the greater difficulty of oxidizing an element to the right of Cr to its group oxidation state is that the air oxidation of MnO_2 in molten potassium hydroxide does not take Mn to its group oxidation state but instead yields the deep green compound potassium manganate(VI), K_2MnO_4. The disproportionation of MnO_4^{2-} in acidic aqueous solution yields manganese(IV) oxide (MnO_2) and the deep purple manganate(VII) ion MnO_4^-:

$$3\,MnO_4^{2-}(aq) + 4\,H^+(aq) \rightarrow 2\,MnO_4^-(aq) + MnO_2(s) + 2\,H_2O(l)$$

(b) Intermediate oxidation states in the 3d series

Key point: Oxidation state +3 is common to the left of the 3d series and +2 is common for metals from the middle to the right of the block.

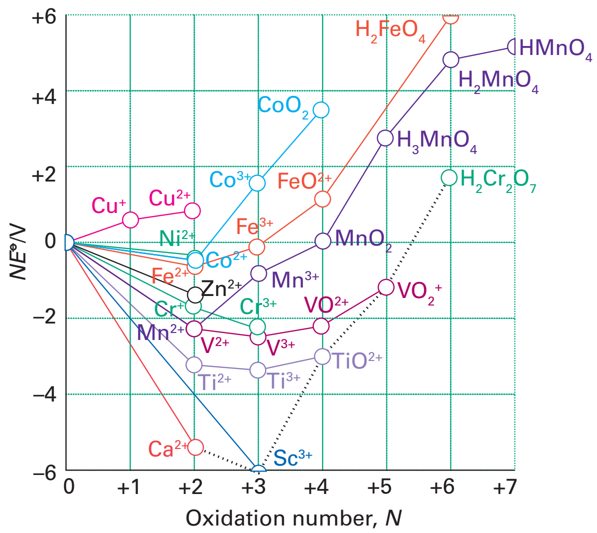


Figure 19.3 A Frost diagram for the first series of the d-block elements in acidic solution (pH = 0). The broken line connects species in their group oxidation states.

Most unipositive d-metal ions (M^+) disproportionate (to M and M^{2+}) because the bonds in the solid metal are so strong and, for the 3d metals, the +2 oxidation state is generally the lowest one it is necessary to consider in aqueous solution. Exceptions are mainly confined to metal–metal bonded and organometallic compounds, such as the metal carbonyls $Ni(CO)_4$ and $Mo(CO)_6$ discussed in Chapter 22, in which the metal is in oxidation state 0. Figure 19.4 shows the second and third ionization energies of the 3d metals, and we can see the expected increase across the period, in line with increasing nuclear charge. The anomalous values for manganese and iron are a result of the very stable d^5 configurations of the Mn^{2+} and Fe^{3+} ions.

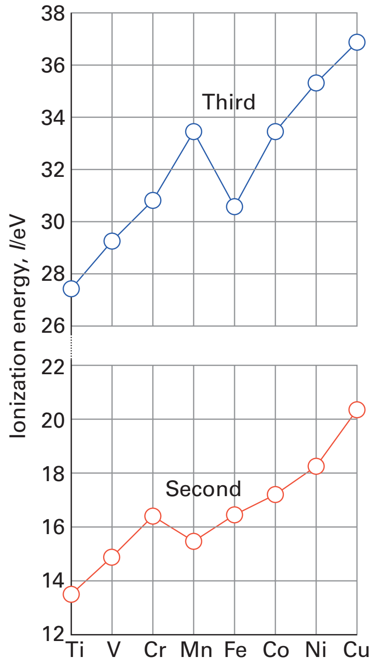


Figure 19.4 The second (red) and third (blue) ionization energies of the 3d metals.

The dipositive aqua ions, $M^{2+}(aq)$ (specifically, the octahedral complexes $[M(OH_2)_6]^{2+}$), play an important role in the chemistry of the 3d-series metals. Many of these ions are coloured as a result of d–d transitions in the visible region of the spectrum (Section 20.4). For example, $Cr^{2+}(aq)$ is blue, $Fe^{2+}(aq)$ is green, $Co^{2+}(aq)$ is pink, $Ni^{2+}(aq)$ is green, and $Cu^{2+}(aq)$ is blue.

The +3 oxidation state is common at the left of the period, and is the only oxidation state normally encountered for scandium. Careful inspection of the Frost diagrams in Fig. 19.3 gives the balance of stability between the +3 and +2 oxidation states. Titanium, vanadium, and chromium all form a wide range of compounds in oxidation state +3, and under normal conditions the +3 oxidation state is more stable than the +2 state. Manganese(II) is especially stable due to its half-filled d shell, and relatively few Mn(III) compounds are known. Beyond manganese, many Fe(III) complexes are known, but are often oxidizing. In acid solution, $Co^{3+}(aq)$ is powerfully oxidizing and O_2 is evolved:

$$4\,Co^{3+}(aq) + 2\,H_2O(l) \rightarrow 4\,Co^{2+}(aq) + 4\,H^+(aq) + O_2 \qquad E_{cell}^{\ominus} = +0.58\ V$$

Species such as Co(III) are stabilized as oxido compounds (for instance, CoO(OH)), which are formed under basic conditions (Section 19.8), or when complexed by good donor ligands such as amines. Aqua ions of Ni^{3+} and Cu^{3+} have not been prepared.

In contrast, M(II) becomes increasingly common from left to right across the series. For example, among the early members of the 3d series, $Sc^{2+}(aq)$ (Group 3) is unknown and $Ti^{2+}(aq)$ (Group 4) is formed only upon bombarding solutions of Ti^{3+} with electrons in the technique known as **pulse radiolysis**. For Groups 5 and 6, $V^{2+}(aq)$ and $Cr^{2+}(aq)$ are thermodynamically unstable with respect to oxidation by H^+ ions:

$$2\,V^{2+}(aq) + 2\,H^+(aq) \rightarrow 2\,V^{3+}(aq) + H_2(g) \qquad E_{cell}^{\ominus} = +0.26\ V$$

Despite this instability, the slowness of the evolution of H_2 makes it possible to work with aqueous solutions of these dipositive ions in the absence of air, and as a result they are useful reducing agents. Beyond Cr (for Mn^{2+}, Fe^{2+}, Co^{2+}, Ni^{2+}, and Cu^{2+}), M(II) is stable with respect to reaction with water, and only Fe^{2+} is oxidized by air.

Water is incompatible with many metal ions because it can serve as an oxidizing agent. A wider range of M^{2+} ions is therefore known in solids than in aqueous solution. For example, $TiCl_2$ can be prepared by the reduction of $TiCl_4$ with hexamethyldisilane,

$$(CH_3)_3Si{-}Si(CH_3)_3(l) + TiCl_4(l) \rightarrow TiCl_2(s) + 2\,(CH_3)_3SiCl(l)$$

but it is oxidized by both water and air to give TiO_2. With the exception of $ScCl_2$, these dihalides are well described by structures containing isolated M^{2+} ions (Table 19.2). The fluorides have a rutile structure (Section 3.9) and most of the heavier halides are layered;

Table 19.2 Structures of the d-metal dihalides*

	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
F	R	R†	R†	R	R	R	R†	L
Cl	L	L	R†	L	L	L	L	L
Br	L	L	L	L	L	L	L	L
I	L	L	L	L	L	L	L	

* R = rutile, L = layered (CdI_2, $CdCl_2$ or related structures).
† The structure is distorted from ideal.
Adapted from A.F. Wells, *Structural inorganic chemistry*. Oxford University Press (1984).

in both cases the metal is in an octahedral site. This shift in structure is understandable because the rutile structure type is associated with ionic bonding and the layered structures are associated with more covalent bonding.

EXAMPLE 19.1 Judging trends in redox stability in the d block

On the basis of trends in the properties of the 3d-series elements, suggest possible M^{2+} aqua ions for use as reducing agents and write a balanced chemical equation for the reaction of one of these ions with O_2 in acidic solution.

Answer We need to identify an element that has an accessible, but oxidizable, M(II) oxidation state. The M(II) state is most stable for the late 3d-series elements with ions of the metals on the left of the series, such as V^{2+}(aq) and Cr^{2+}(aq) being too strongly reducing to be used easily in water. By contrast, the Fe^{2+}(aq) ion is only weakly reducing and the Co^{2+}(aq), Ni^{2+}(aq), and Cu^{2+}(aq) ions not reducing in water. The Latimer diagram for iron indicates that Fe^{3+} is the only accessible higher oxidation state of iron in acidic solution:

$$Fe^{3+} \xrightarrow{+0.77} Fe^{2+} \xrightarrow{-0.44} Fe$$

The chemical equation for the oxidation is then

$$4\,Fe^{2+}(aq) + O_2(g) + 4\,H^+(aq) \rightarrow 4\,Fe^{3+}(aq) + 2\,H_2O(l)$$

Self-test 19.1 Refer to the appropriate Latimer diagram in *Resource section* 3 and identify the oxidation state and formula of the species that is thermodynamically favoured when an acidic aqueous solution of V^{2+} is exposed to oxygen.

(c) Intermediate oxidation states in the 4d and 5d series

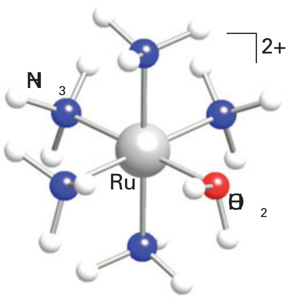
Key points: Complexes of M(II) with σ-donor ligands are common for the 3d-series metals but complexes of M(II) of the 4d- and 5d-series metals are less common; they generally contain π-acceptor ligands.

In contrast to the 3d-series, the 4d- and 5d-series metals only rarely form simple M^{2+}(aq) ions. A few examples have been characterized, including $[Ru(OH_2)_6]^{2+}$, $[Pd(OH_2)_4]^{2+}$, and $[Pt(OH_2)_4]^{2+}$. However, the 4d- and 5d-series metals do form many M(II) complexes with ligands other than H_2O; they include the very stable d^6 octahedral complexes, such as (**1**), and the much rarer square-pyramidal d^6 complexes, such as (**2**), which form with bulky ligands. Palladium(II) and platinum(II) form many square-planar d^8 complexes, such as $[PtCl_4]^{2-}$. Much of the rationalization of this behaviour relates to the bonding of the ligands, and cannot be fully understood without reference to the concepts introduced in Chapter 20; however, examples of complexes are included here for completeness.

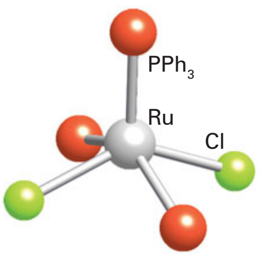
The Ru(II) complex in (**1**) is obtained by reducing $RuCl_3.3H_2O$ with zinc in the presence of ammonia; it is a useful starting material for the synthesis of a range of ruthenium(II) pentaammine complexes that have π-acceptor ligands, such as CO, as the sixth ligand:

$$[Ru(NH_3)_5(OH_2)]^{2+}(aq) + L(aq) \rightarrow [Ru(NH_3)_5L]^{2+}(aq) + H_2O(l) \qquad (L = CO, N_2, N_2O)$$

These Ru and the related Os pentaammine species are strong π donors, and consequently the complexes they form with the π acceptors CO and N_2 are stable.



1 $[Ru(OH_2)(NH_3)_5]^{2+}$



2 $[Ru(Cl)_2(PPh_3)_3]$

19.4 Oxidation states down a group

Key points: In Groups 4–10, the highest oxidation state of an element becomes more stable on descending a group, with the greatest change in stability occurring between the first two rows of the d block; ease of oxidation of the metal does not correlate with the highest available oxidation state.

The stability of an oxidation state of an element changes on descending a group. Figure 19.5 is the Frost diagram for the chromium group: note that Mo(VI) and W(VI) lie below Cr(VI) in $H_2Cr_2O_7$, indicating that the maximum oxidation state is more stable for Mo and W, with the greatest change in stability occurring between Cr and Mo. This pattern is repeated

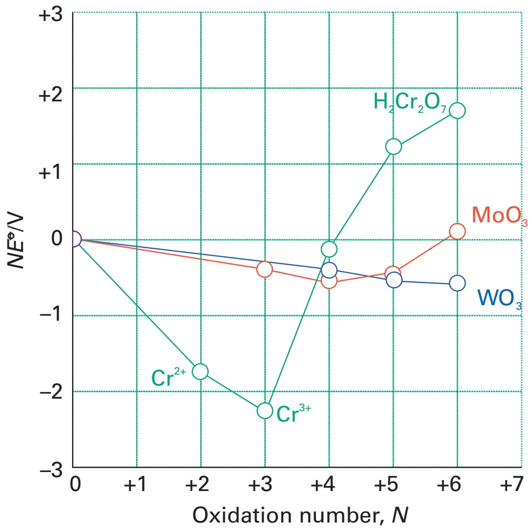


Figure 19.5 A Frost diagram for the chromium group in the d block (Group 6) in acidic solution (pH = 0).

across the d block: for Groups 4–10 the highest oxidation state of an element becomes more stable on descending a group, with the greatest change in stability occurring between the first two series of the block.

The increasing stability of high oxidation states for the heavier d metals can be seen in the formulas of their halides (Table 19.3), and the limiting formulas MnF_4, TcF_6, and ReF_7 show a greater ease of oxidizing the 4d- and 5d-series metals than the 3d-series. The hexafluorides of the heavier d metals (as in PtF_6) have been prepared from Group 6 through to Group 10 except for Pd. In keeping with the stability of high oxidation states for the heavier metals, WF_6 is not a significant oxidizing agent. However, the oxidizing character of the hexafluorides increases to the right, and PtF_6 is so potent that it can oxidize O_2 to O_2^+:

$$O_2(g) + PtF_6(s) \rightarrow (O_2)PtF_6(s)$$

Even Xe can be oxidized by PtF_6 (Section 18.5).

The ability to achieve the highest oxidation state does not correlate with the ease of oxidation of the bulk metal to an intermediate oxidation state. For example, although elemental iron is susceptible to oxidation by $H^+(aq)$ under standard conditions,

$$Fe(s) + 2H^+(aq) \rightarrow Fe^{2+}(aq) + H_2(g) \qquad E_{cell}^{\ominus} = +0.44\ V$$

no chemical oxidizing agent has been found that will take it to its group oxidation state of +8 in solution. By contrast, although the two heavier metals in Group 8 (Ru and Os) are not oxidized by H^+ ions in acidic aqueous solution:

$$Os(s) + 2H_2O(aq) \rightarrow OsO_2(s) + 2H_2(g) \qquad E_{cell}^{\ominus} = -0.65\ V$$

they can be oxidized by oxygen to the +8 state, as RuO_4 and OsO_4:

$$Os(s) + 2O_2(g) \rightarrow OsO_4(s)$$

Table 19.3 Highest oxidation states of the d-block binary halides*

Group 4	5	6	7	8	9	10	11
TiI_4	VF_5	CrF_5†	MnF_4	$FeBr_3$	CoF_4	NiF_4	$CuBr_2$
ZrI_4	NbI_5	$MoCl_6$	$TcCl_6$	RuF_6	RhF_6	PdF_4	AgF_3
HfI_4	TaI_5	WBr_6	ReF_7	OsF_6	IrF_6	PtF_6	AuF_5

* The formulas show the least electronegative halide that brings out the highest oxidation state of the d metal.
† CrF_6 exists for several days at room temperature in a passivated Monel chamber.

We see from Fig. 19.5 that the Frost diagrams for the positive oxidation states of Mo and (especially) W are quite flat. This flatness indicates that neither element exhibits the marked tendency to form M(III) that is so characteristic of chromium. Mononuclear complexes of Mo and W in oxidation states +3, +4, +5, and +6 are common.

19.5 Structural trends

Key points: The 4d- and 5d-series elements often exhibit higher coordination numbers than their 3d-series congeners; compounds of d-metals in high oxidation states tend to have covalent structures.

We might expect the ionic radii of d-metal ions to follow the same pattern as atomic radii and decrease across each period. However, in addition to this general trend, there are some subtle effects on the size of the ions caused by the order in which d orbitals are occupied and which will be explained more fully in Chapter 20. Figure 19.6 shows the variation in radius of the M^{2+} ions for six-coordinate complexes of the 3d-series metals. To understand the two trends shown in the illustration we need to know that three of the 3d orbitals point between the ligands and that the remaining two point directly at them (this feature is explained more fully in Section 20.1). For the so-called 'low-spin complexes', in which electrons first individually occupy the three 3d orbitals that point between the ligands, there is a general decrease in radius across the series up to the d^6 ion Fe^{2+}. After Fe^{2+}, the additional electrons occupy the two d orbitals that point towards the ligands, which they repel slightly and thus result in an effective increase in radius. The trend for the so-called 'high-spin complexes', in which the electrons occupy all five 3d orbitals singly before pairing with any already present, is more complicated. Initially, at Ti^{2+} (d^2) and V^{2+} (d^3), the electrons occupy the three 3d orbitals that point between the ligands and the radius decreases. The next two electrons occupy the two orbitals that point at the ligands, and the radii increase accordingly. Then the sequence starts again at Fe^{2+} (d^6), as the additional electrons pair with those already present, first in the 'non-repelling' set of three orbitals and then, finally, in the two 'repelling' orbitals.

As might be anticipated from consideration of atomic and ionic radii, the 4d- and 5d-series elements often have higher coordination numbers than their smaller 3d-series congeners. Table 19.4 illustrates this trend for the fluorido and cyanido complexes of the early d metals. Note that with the small F^- ligand, these 3d-series metals tend to form six-coordinate complexes but that the larger 4d- and 5d-series metals in the same oxidation state tend to form seven-, eight-, and nine-coordinate complexes. The octacyanidomolybdate complex, $[Mo(CN)_8]^{3-}$, illustrates the tendency towards high coordination numbers with compact ligands. The same complex is readily reduced electrochemically or chemically without change of coordination number:

$$[Mo(CN)_8]^{3-}(aq) + e^- \rightarrow [Mo(CN)_8]^{4-}(aq) \qquad E^{\ominus} = +0.73\ V$$

Structural changes also result from changes in metal oxidation state. Compounds of d metals in low oxidation states often exist as ionic solids, whereas compounds of d metals in high oxidation states tend to take on covalent character: compare OsO_2, which is an ionic solid with the rutile structure, and OsO_4, which is a covalent molecular species (Section 19.8). We discussed the effect in Section 1.9e.

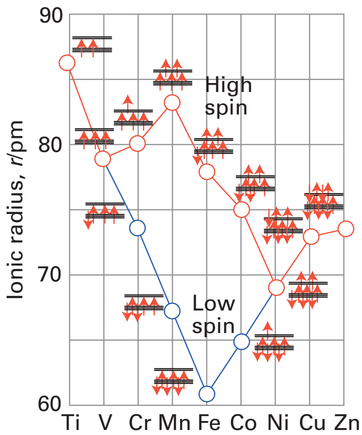


Figure 19.6 The ionic radii of the M^{2+} ions of the 3d metals. Where there are alternatives, red represents high-spin and blue low-spin complexes. In the orbital energy diagrams, the three lower levels are the d orbitals that point between ligands and the two upper levels are the d orbitals that point directly at them.

Table 19.4 Coordination numbers of some early d-block fluorido and cyanido complexes*

	Group 3	4	5
3d	$(NH_4)_3[ScF_6]$ (6)	$Na_2[TiF_6]$ (6)	$K[VF_6]$ (6); $K_2[V(CN)_7].2H_2O$ (7)
4d	$Na_6[YF_9]$ (9)	$Na_3[ZrF_7]$ (7)	$K_2[NbF_7]$ (7); $K_4[Nb(CN)_8]$ (8)
5d	$Na_6[LaF_9]$ (9)	$Na_3[HfF_7]$ (7)	$K_3[TaF_8]$ (8)

* The number in parentheses is the coordination number of the d-metal atom in the complex anion enclosed in square brackets.

19.6 Noble character

Key point: Metals on the right of the d block tend to exist in low oxidation states and form compounds with soft ligands.

With the exception of Group 12, the metals at the lower right of the d block are resistant to oxidation. This resistance is largely due to strong intermetallic bonding and high ionization energies. It is most evident for Ag, Au, and the 4d- and 5d-series metals in Groups 8–10 (Fig. 19.7). The latter are referred to as the **platinum metals** because they occur together in platinum-bearing ores. In recognition of their traditional use, Cu, Ag, and Au are referred to as the **coinage metals**. Gold occurs as the metal; silver, gold, and the platinum metals are also recovered in the electrolytic refining of copper. The prices of the individual platinum metals vary widely because they are recovered together but their consumption is not proportional to their abundance. Rhodium is by far the most expensive metal in this group because it is widely used in industrial catalytic processes and in automotive catalytic converters (Box 26.1).

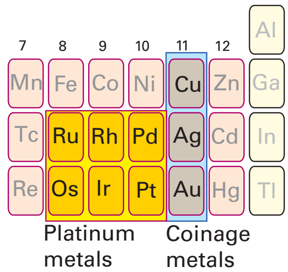


Figure 19.7 The location of the platinum and coinage metals in the periodic table.

Copper, silver, and gold are not susceptible to oxidation by H^+ under standard conditions, and this noble character accounts for their use, together with platinum, in jewellery and ornaments. Aqua regia, a 3:1 mixture of concentrated hydrochloric and nitric acids, is an old but effective reagent for the oxidation of gold and platinum. Its function is twofold: the NO_3^- ions provide the oxidizing power and the Cl^- ions act as complexing agents. The overall reaction is

$$Au(s) + 4\,H^+(aq) + NO_3^-(aq) + 4\,Cl^-(aq) \rightarrow [AuCl_4]^-(aq) + NO(g) + 2\,H_2O(l)$$

The active species in solution are thought to be Cl_2 and NOCl, which are generated in the reaction

$$3\,HCl(aq) + HNO_3(aq) \rightarrow Cl_2(aq) + NOCl(aq) + 2\,H_2O(l)$$

Oxidation state preferences are erratic in Group 11. For Cu, the +1 and +2 states are most common, but for Ag +1 is typical, and for Au +1 and +3 are common. The simple aqua ions $Cu^+(aq)$ and $Au^+(aq)$ undergo disproportionation in aqueous solution:

$$2\,Cu^+(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$$

$$3\,Au^+(aq) \rightarrow 2\,Au(s) + Au^{3+}(aq)$$

Complexes of Cu(I), Ag(I), and Au(I) are often linear. For example, $[H_3NAgNH_3]^+$ forms in aqueous solution and linear $[XAgX]^-$ complexes have been identified by X-ray crystallography. The currently preferred explanation for the tendency towards linear coordination is the similarity in energy of the outer nd and $(n+1)$s and $(n+1)$p orbitals, which permits the formation of collinear spd hybrids (Fig. 19.8).

The soft Lewis acid character of Cu^+, Ag^+, and Au^+ is illustrated by their affinity order, which is $I^- > Br^- > Cl^-$. Complex formation, as in the formation of $[Cu(NH_3)_2]^+$ and $[AuI_2]^-$, provides a means of stabilizing the +1 oxidation state of these metals in aqueous solution. Many tetrahedral complexes are also known for Cu(I), Ag(I), and Au(I) (Section 7.8).

Square-planar complexes are common for the platinum metals and gold in oxidation states that yield the d^8 electronic configuration, which include Rh(I), Ir(I), Pd(II), Pt(II), and Au(III) (Section 20.1f). An example is $[Pt(NH_3)_4]^{2+}$. Characteristic reactions for these

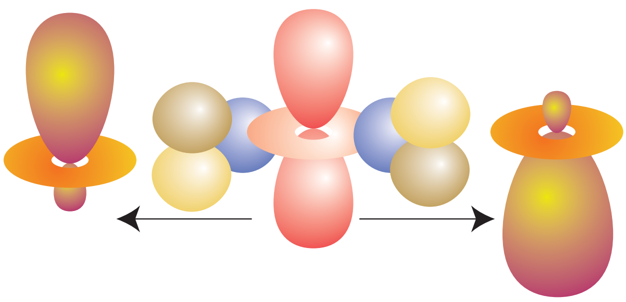


Figure 19.8 The hybridization of s, p_z, and d_{z^2} with the choice of phases shown here produces a pair of collinear orbitals that can be used to form strong σ bonds.

complexes are ligand substitution (Section 21.3) and, except for Au(III) complexes, oxidative addition (Section 22.22).

The noble character that these elements developed across the d block is suddenly lost at Group 12 (Zn, Cd, Hg), where the metals once again become susceptible to atmospheric oxidation. The greater ease of oxidation of the Group 12 metals is due to a reduction in the extent of intermetallic bonding and an abrupt lowering of d-orbital energies at the end of the d block, with the higher energy $(n+1)$s electrons participating in reactions.

Representative compounds

As a result of the convention of assigning a negative oxidation number to any nonmetal in combination with a metal, high formal oxidation states for d metals may be encountered in their compounds, such as Re(VII) in $[ReH_9]^{2-}$ and W(VI) in $W(CH_3)_6$. However, these compounds are not oxidizing agents in the usual sense, and are best discussed together with other organometallic compounds (Chapter 22). The discussion here is confined to compounds containing electronegative ligands such as the halogens, oxygen, nitrogen, and sulfur.

19.7 Metal halides

Key points: Binary halides of the d-block elements span all metals and most oxidation states; dihalides are typically ionic solids, with higher halides taking on covalent character.

Binary metal halides of the d-block elements occur for all the elements with nearly all oxidation states represented. As we should expect, the more strongly oxidizing halogens bring out the higher oxidation states, with the corollary that the low oxidation state binary halides are more stable as iodides and bromides.

Of all the groups, only the members of Group 11 (Cu, Ag, Au) have simple monohalides. For Cu, these salts are highly insoluble in water and dissolve only when complexed by other ligands (which include halides). Silver(I) halides are sparingly soluble and photosensitive, decomposing to the metal (a process that is exploited in photography). The only monohalide of Au that exists is the chloride; it is oxidized by water.

More common are the dihalides which, as noted in Section 19.3b (Table 19.2), are typically ionic solids that dissolve in water to give M^{2+}(aq) ions; some of the earlier dihalides are reducing. Many of the intermediate halides exhibit metal–metal bonds and form clusters (Section 19.11).

Higher halides exist for most of the d block, and covalent character becomes more prevalent with high oxidation state, especially for the lower halogens. For instance, in Group 4, whereas TiF_4 is a solid with melting point 284°C, $TiCl_4$ melts at −24°C and boils at 136°C. In Group 6, not even the fluoride has ionic character and both MoF_6 and WF_6 are liquids at room temperature.

19.8 Metal oxides and oxido complexes

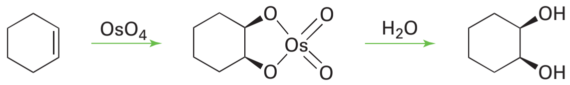
Because oxygen is readily available in an aqueous environment, in the atmosphere, and as the donor atom in many organic molecules, it is not surprising that oxides and oxygen-containing ligands play a major role in the chemistry of the metallic elements.

(a) Metal oxides

Key point: Many different oxides of the d-block elements exist, with a wide variety of structures, varying from ionic lattices to covalent molecules.

Many different oxides are known for the d-block elements, with a number of different structures. We have already noted the ability of oxygen to bring out the highest oxidation state for some elements, but oxides exist for some elements in very low oxidation states: in Cu_2O, copper is present as Cu(I). Monoxides are known for all of the 3d-series metals, except Cr. The monoxides have the rock-salt structure characteristic of ionic solids but their properties, which are discussed in more detail in Chapter 24, indicate significant deviations from the simple ionic $M^{2+}O^{2-}$ model. For example, TiO has metallic conductivity and FeO is always deficient in iron. The early d-block monoxides are strong reducing agents. Thus, TiO is easily oxidized by water or oxygen, and MnO is a convenient oxygen scavenger that is used in the laboratory to remove oxygen impurity in inert gases down to the parts-per-billion range.

As we have already noted, very high oxidation state oxides can show covalent structures. For example ruthenium tetroxide and osmium tetroxide are low melting, highly volatile, toxic, molecular compounds that are used as selective oxidizing agents. Indeed, osmium tetroxide is used as the standard reagent to oxidize alkenes to *cis*-diols:



(b) Mononuclear oxido complexes

Key points: The conversion of an aqua ligand to an oxido ligand is favoured by a high pH and by a high oxidation state of the central metal atom; in vanadium complexes in oxidation state +4 or +5, the site *trans* to the oxido ligand may be vacant or occupied by a weakly coordinating ligand.

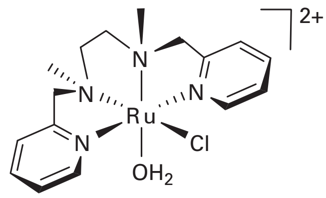
Our focus in this section is on the oxido ligand (O^{2-}) and its ability to bring out the high oxidation states of the chemically hard metals on the left of the d block. Another important issue is the relation between oxido and aqua (H_2O) ligands resulting from proton transfer equilibria.

Elements in high oxidation states typically occur as oxoanions in aqueous solution, such as MnO_4^-, which contains Mn(VII), and CrO_4^{2-}, which contains Cr(VI). The existence of these oxoanions contrasts with the existence of simple aqua ions for the same metals in lower oxidation states, such as $[Mn(OH_2)_6]^{2+}$ for manganese(II) and $[Cr(OH_2)_6]^{3+}$ for chromium(III).

The formation of an oxido complex rather than an aqua complex is favoured by high pH because the OH^- ions in basic solution tend to remove protons from the aqua ligands. The occurrence of aqua complexes for low oxidation state metal cations is explained by noting the relatively small electron-withdrawing effect exerted by these cations on the O atoms of the OH_2 ligand. As a result, the aqua ligands are only weak proton donors. A metal ion in a high oxidation state, however, depletes the electron density on the O atoms attached to it and thereby increases the Brønsted acidity of H_2O and OH^- ligands.

The influence of pH on the stabilities of different oxidation states is best summarized by a Pourbaix diagram (Section 5.14). The example shown in Fig. 19.9 is for a complex (**3**) containing a tetradentate ligand L, which results in a similar coordination geometry for a wide range of conditions. The aqua complex *cis*-$[RuLCl(OH_2)]^+$ in solution at pH = 2 is stable up to +0.40 V. Just above this potential, simple oxidation (that is, electron removal) occurs to give *cis*-$[RuLCl(OH_2)]^{2+}$. Under even more strongly oxidizing conditions (at +0.95 V) and pH = 2, both oxidation and deprotonation occur and result in the formation of a Ru(IV) oxido species, *cis*-$[RuLCl(O)]^+$. At an even higher potential (about +1.4 V), further oxidation yields *cis*-$[RuLCl(O)]^{2+}$. As already remarked, the influence of more basic conditions is to deprotonate an OH_2 ligand and thus to favour hydroxido or oxido complexes. For example, at pH = 8, the Ru(III) species is deprotonated to the hydroxido complex *cis*-$[RuLCl(OH)]^+$, which in turn is converted to the oxido-Ru(IV) complex at a lower potential than for the Ru(III)–Ru(IV) transformation at pH = 2.

Table 19.5 gives a list of simple complexes containing oxido ligands. Many complexes are known that contain the vanadium(IV) group, VO^{2+} (sometimes referred to as vanadyl), with V in its penultimate oxidation state (+4). Vanadyl complexes generally contain four additional ligands and are square pyramidal (**4**). Many of these d^1 complexes are blue (as a result of a d–d transition) and may take on a weakly bound sixth ligand *trans* to the oxido ligand. The vanadyl VO bond length (158 pm) in $[VO(acac)_2]$ is short compared with the four VO bond lengths to the acac ligand (197 pm). The short bond lengths in vanadyl complexes, together with high VO stretching wavenumbers (940–1000 cm^{-1}) provide strong evidence for VO multiple bonding in which lone pairs on the oxido ligand are donated to the central V atom. The V3d–O2p π-orbital overlap involved in a multiple bond is illustrated in Fig. 19.10; thus it can be seen that the O^{2-} ligand is not only a σ donor but also a π donor capable of making two π bonds and resulting in a bond order of up to three (though by convention such interactions are depicted as M=O, preserving electroneutrality). This strong



3 *cis*-$[RuLCl(OH_2)]^{2+}$

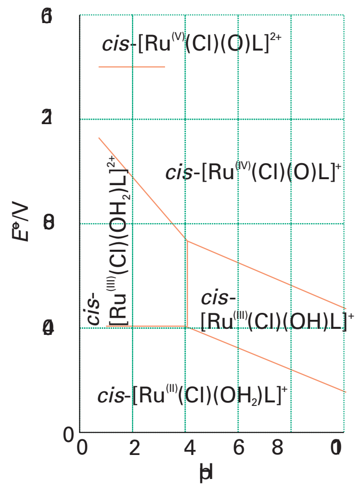
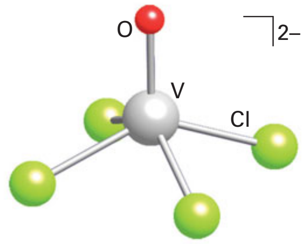


Figure 19.9 A Pourbaix diagram for *cis*-$[RuLCl(OH_2)]^{2+}$ (**3**) and related species. (Adapted from C.-K. Li, W.-T. Wang, C.-M. Chi, K.-Y. Wong, R.-J. Wang, and T.C.W. Mak, *J. Chem. Soc., Dalton Trans.* 1991, 1909.)



4 $[VOCl_4]^{2-}$

Table 19.5 Some common monoxido and dioxido complexes

Group	Element, configuration	Structure	Formula
5	V(IV), d^1	Square pyramidal	$[V(O)(acac)_2]$, $[V(O)Cl_4]^{2-}$
	V(V), d^0	*cis*-octahedral	$[V(O)_2(OH_2)_4]^+$
6	Mo(VI), d^0; W(VI), d^0	Tetrahedral	$[M(O)_2(Cl)_2]$
		cis-octahedral	$[Mo(O)_2(acac)_2]$
7, 8	Re(V), d^2; Os(VI), d^2	*trans*-octahedral	$[Re(O)_2(CN)_4]^{3-}$
			$[Os(O)_2(Cl)_2]$
			$[Os(O)_2(Cl)_4]^{2-}$

multiple bonding with the O atom appears to be responsible for the *trans* influence of the oxido ligand, which disfavours attachment of the ligand *trans* to O (Section 21.4).

Vanadium in its highest (and group) oxidation state, +5, forms an extensive series of oxido compounds, many of which are the polyoxo species discussed later. The simplest oxido complex $[V(O)_2(OH_2)_4]^+$ exists in the acidic solution formed when the sparingly soluble vanadium(V) oxide, V_2O_5, dissolves in water. This pale yellow complex has a *cis* geometry (**5**). Here again we see the *trans* influence of an oxido ligand: the *cis* geometry minimizes competition between the oxido ligands for π-bonding to the V atom. As shown in Fig. 19.11, the two O^{2-} ligands in the *trans* configuration can π-bond with the same two d orbitals; in the *cis* configuration the metal atom has only one d orbital in common with the π orbitals on the O atoms.

In contrast to the *cis* structure of the d^0 complex $[V(O)_2(OH_2)_4]^+$, many *trans*-dioxido complexes are known for the d^2 metal centres Os(VI) and Re(V) (**6**); Table 19.5 gives some examples. This geometry is probably favoured because the *trans* configuration leaves a vacant low-energy orbital that can be occupied by the two d electrons. According to this explanation, the avoidance of the destabilizing effect of the d^2 electrons more than offsets the disadvantage of the ligands having to compete for the same d orbital in the *trans*-oxido geometry.

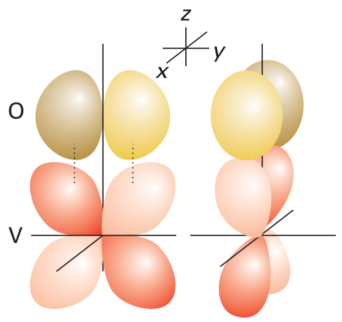
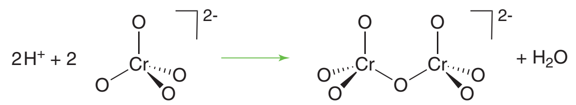


Figure 19.10 The d−p π-bonding between O and V in the vanadyl group VO^{2+}. It is convenient to think of the O as having oxidation number −2 and V as +4. Thus, both electron pairs for the two π bonds are donated from the O^{2-} ion into the metal d_{yz} and d_{zx} orbitals.

(c) Polyoxometallates

Key points: In their highest oxidation states, metals in Groups 5 and 6 readily form polyoxometallates and heteropolyoxometallates; pH is crucial in determining which compounds form.

A **polyoxometallate** is an oxoanion containing more than one metal atom. The H_2O ligand created by protonation of an oxido ligand at low pH can be eliminated from the central metal atom and thus lead to the condensation of mononuclear oxometallates. A familiar example is the reaction of a basic chromate solution, which is yellow, with excess acid to form the oxido-bridged dichromate ion, which is orange:



$$2\,CrO_4^{2-}(aq) + 2\,H^+(aq) \rightarrow Cr_2O_7^{2-}(aq) + H_2O(l)$$

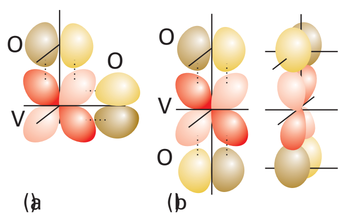
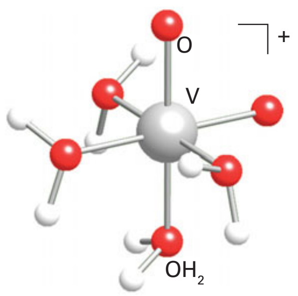
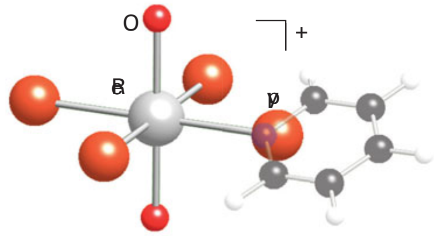


Figure 19.11 Comparison of the competition between *cis* and *trans* oxygens attached to vanadium. (a) In the *cis* configuration the metal has only one d orbital in common with the π orbitals on the O atoms. (b) In the *trans* configuration the metal atom has two d orbitals in common with the π orbitals on the two O atoms.



5 $[V(O)_2(OH_2)_4]^+$



6 $[Re(O)_2(py)_4]^+$, py = C_5H_5N

In highly acidic solution, oxido-bridged Cr(VI) species with longer chains are formed. The tendency for Cr(VI) to form polyoxo species is limited by the fact that the O tetrahedra link only through vertices: edge and face bridging would result in too close an approach of the metal atoms. By contrast, it is found that five- and six-coordinate metal oxido complexes, which are common with the larger 4d- and 5d-series metal atoms, can share oxido ligands between either vertices or edges. These structural possibilities lead to a richer variety of polyoxometallates than found with the 3d-series metals.

Chromium's neighbours in Groups 5 and 6 form six-coordinate polyoxo complexes (Fig. 19.12). In Group 5, the polyoxometallates are most numerous for vanadium, which forms many V(V) complexes and a few V(IV) or mixed oxidation state V(IV)−V(V) polyoxido complexes. Polyoxometallate formation is most pronounced in Groups 5 and 6 for V(V), Mo(VI), and W(VI).

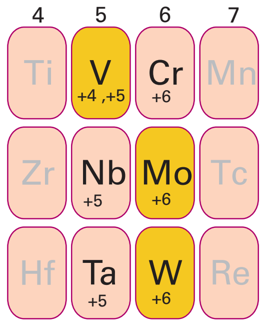
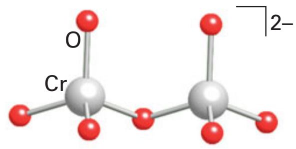
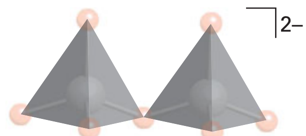


Figure 19.12 Elements in the d block that form polyoxometallates. The elements coloured yellow form the greatest variety of polyoxometallates.

It is often convenient to represent the structures of the polyoxometallate ions by polyhedra, with the metal atom understood to be in the centre and O atoms at the vertices. For example, the sharing of O atom vertices in the dichromate ion, $Cr_2O_7^{2-}$, may be depicted in either the traditional way (**7**) or in the polyhedral representation (**8**). Similarly, the important M_6O_{19} structure of $[Nb_6O_{19}]^{8-}$, $[Ta_6O_{19}]^{8-}$, $[Mo_6O_{19}]^{2-}$, and $[W_6O_{19}]^{2-}$ is depicted by the conventional or polyhedral structures shown in Fig. 19.13. The structures for this series of polyoxometallates contain terminal O atoms (those projecting out from a single metal atom) and two types of bridging O atoms: two-metal bridges, M−O−M, and one hypercoordinated O atom in the centre of the structure that is common to all six metal atoms. The structure consists of six MO_6 octahedra, each sharing an edge with four neighbours. The overall symmetry of the M_6O_{19} array is O_h. Another example of a polyoxometallate is $[W_{12}O_{40}(OH)_2]^{10-}$ (**9**). As shown by this formula, the polyoxoanion is partially protonated.



7 $Cr_2O_7^{2-}$



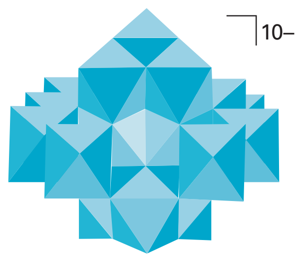
8 $Cr_2O_7^{2-}$

Polyoxometallate anions can be prepared by carefully adjusting pH and concentrations, for example polyoxomolybdates and polyoxotungstates are formed by acidification of solutions of the simple molybdate or tungstate:

$$6\,[MoO_4]^{2-}(aq) + 10\,H^+(aq) \rightarrow [Mo_6O_{19}]^{2-}(aq) + 5\,H_2O(l)$$

$$8\,[MoO_4]^{2-}(aq) + 12\,H^+(aq) \rightarrow [Mo_8O_{26}]^{4-}(aq) + 6\,H_2O(l)$$

Mixed metal polyoxometallates are also common, as in $MoV_9O_{28}^{5-}$, and there is a large class of heteropolyoxometallates, such as the molybdates and tungstates, that also incorporate P, As, and other heteroatoms. For example, $[PMo_{12}O_{40}]^{3-}$ contains a PO_4^{3-} tetrahedron that shares O atoms with surrounding octahedral MoO_6 groups (**10**). Many different heteroatoms can be incorporated into this structure, and the general formulation is $[X(+N)Mo_{12}O_{40}]^{(8-N)-}$ where $X(+N)$ represents the oxidation state of the heteroatom X, which may be As(V), Si(IV), Ge(IV), or Ti(IV). An even broader range of heteroatoms is observed with the analogous tungsten heteropolyoxoanions. Heteropolyoxomolybdates and tungstates can undergo one-electron reduction with no change in structure but with the formation of a deep blue colour. The colour seems to arise from the excitation of the added electron from Mo(V) or W(V) to an adjacent Mo(VI) or W(VI) site.



9 $[W_{12}O_{40}(OH)_2]^{10-}$

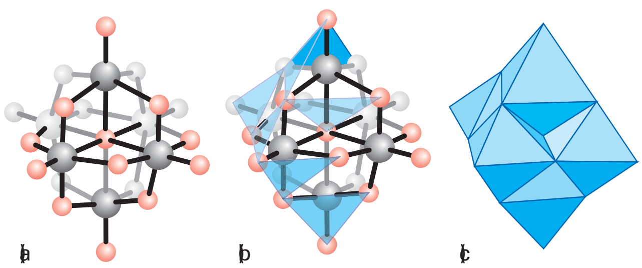
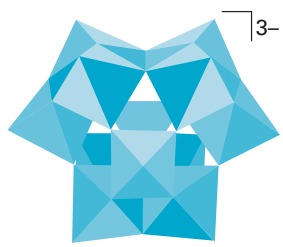


Figure 19.13 (a) Conventional and (c) polyhedral representation of the six edge-shared octahedra as found in $[M_6O_{19}]^{2-}$. The intermediate structure (b) shows the process of constructing the polyhedral representation in progress.



10 $[PMo_{12}O_{40}]^{3-}$

19.9 Metal sulfides and sulfide complexes

Sulfur is softer as a Lewis base and less electronegative than oxygen; it therefore has a broader range of oxidation states and a stronger affinity for the softer metals on the right of the d block. For example, zinc(II) sulfide is readily precipitated when $Zn^{2+}(aq)$ is added to an aqueous solution containing H_2S and NH_3 (to adjust the pH) whereas the harder Lewis acid $Sc^{3+}(aq)$ gives $Sc(OH)_3(s)$ instead. The covalent contribution to the lattice enthalpy of all the softer metal sulfides cannot be overcome in water, and in general the sulfides are only very sparingly soluble. Another striking feature of sulfur is its tendency to form catenated sulfide ions, Section 16.14. A broad set of compounds containing S_n^{2-} ions can be prepared, and the smaller polysulfides can serve as chelating ligands towards d-metal ions.

(a) Monosulfides

Key point: **Monosulfides with the nickel-arsenide structure are formed by most 3d metals.**

As with the d-metal monoxides, the monosulfides are most common in the 3d series (Table 19.6). In contrast to the monoxides, however, most of the monosulfides have the nickel-arsenide structure (Fig. 3.36). The different structures are consistent with the rock-salt structure of the monoxides being favoured by the ionic (harder) cation–anion combination and the nickel-arsenide structure being favoured by more covalent (softer) combinations. The nickel-arsenide structure is expected when there is appreciable metal–metal bonding, resulting in the shorter metal–metal distances found for the hexagonal packing. Monosulfides are formed by most 3d-series metal ions.

(b) Disulfides

Key points: **The 4d- and 5d-series metals often form disulfides with alternating layers of metal ions and sulfide ions; binary disulfides of the early d metals often have a layered structure, whereas Fe^{2+} and many of the later d-metal disulfides contain discrete S_2^{2-} ions.**

The disulfides of the d metals fall into two broad classes (Table 19.7). One class consists of layered compounds with either the CdI_2 or the MoS_2 structure; the other consists of compounds containing discrete S_2^{2-} groups, the pyrites and marcasite structures.

The layered disulfides are built from a sulfide layer, a metal layer, and then another sulfide layer (for example, Fig. 19.14). These sandwiches stack together in the crystal with sulfide layers in one slab adjacent to a sulfide layer in the next. Clearly, this crystal

Table 19.6 Structures of d-block MS compounds*

	Group 4	5	6	7	8	9	10
Nickel-arsenide structure (shaded)	Ti	V		Mn†	Fe	Co	Ni
Rock-salt structure (unshaded)	Zr	Nb					

* Metal monosulfides of Group 6 are not shown; some of the heavier metals have more complex structures.

† MnS has two polymorphs; one has a rock-salt structure, the other has a wurtzite structure.

Table 19.7 Structures of d-block MS_2 compounds*

	Group 4	5	6	7	8	9	10	11
Layered (shaded)	Ti			Mn	Fe	Co	Ni	Cu
Pyrite or marcasite	Zr	Nb	Mo		Ru	Rh		
(unshaded)	Hf	Ta	W	Re	Os	Ir	Pt	

* Metals not shown do not form disulfides or have disulfides with complex structures.
Adapted from A.F. Wells, *Structural inorganic chemistry*. Oxford University Press (1984).

structure is not consistent with a simple ionic model and its formation is a sign of covalence in the bonds between the soft sulfide ion and d-metal cations. The metal ion in these layered structures is surrounded by six S atoms. Its coordination environment is octahedral in some cases (such as PtS_2, which adopts the CdI_2 structure shown in Fig. 19.14) and trigonal prismatic in others (MoS_2). The layered MoS_2 structure is favoured by S—S bonding as indicated by short S—S distances within each of the MoS_2 slabs. The common occurrence of the trigonal-prismatic structure in many of these compounds is in striking contrast to the isolated metal complexes, where the octahedral arrangement of ligands is by far the most common. Some of the layered metal sulfides readily undergo intercalation reactions in which ions or molecules penetrate between adjacent sulfide layers (Section 24.10).

Compounds containing discrete S_2^{2-} ions adopt the pyrite or marcasite structure (Fig. 19.15). The stability of the formal S_2^{2-} ion in metal sulfides is much greater than that of the O_2^{2-} ion in peroxides, and there are many more metal sulfides in which the anion is S_2^{2-} than there are peroxides.

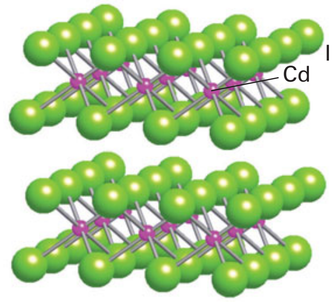


Figure 19.14 The CdI_2 structure adopted by many disulfides.

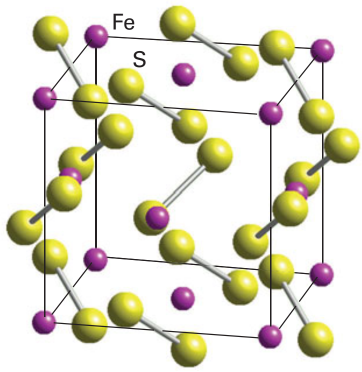


Figure 19.15 The structure of pyrite, FeS_2.

EXAMPLE 19.2 Contrasting the structures of two different d-block disulfides

Compare the structures of MoS_2 and FeS_2, and explain their existence in terms of the oxidation states of the metal ions.

Answer We need to decide whether the metals are likely to be present as M(IV), in which case the two S atoms would be present as S^{2-} ions, or whether the metal is present as M(II) with the S atoms present as the S—S bonded species S_2^{2-}. Because it is readily oxidized, S^{2-} will be found only with a metal ion in an oxidation state that is not easily reduced. As with many of the 4d- and 5d-series metals, Mo is easily oxidized to Mo(IV), therefore Mo(IV) can coexist with S^{2-}. Molybdenum(IV) sulfide (MoS_2) has the layered structure typical of metal disulfides. In contrast, Fe is not readily oxidized to Fe(IV). Therefore, Fe(IV) cannot coexist with S^{2-}. The compound is therefore likely to contain Fe(II) and S_2^{2-}. The mineral name for FeS_2 is pyrite; its common name, 'fool's gold', indicates its misleading colour.

Self-test 19.2 Suggest a use for molybdenum(IV) sulfide that makes use of its solid-state structure. Rationalize your suggestion.

(c) Sulfide complexes

Key point: Chelating polysulfide ligands are common in metal—sulfur coordination compounds of the 4d- and 5d-series metals.

The coordination chemistry of sulfur is quite different from that of oxygen. Much of this difference is connected with the ability of sulfur to catenate (form chains), the availability of low-lying empty 3d orbitals that allow an S atom to act as a π acceptor, and the preference of sulfur for metal centres that are not highly oxidizing (Section 16.11).

Simple thiometallate complexes such as $[MoS_4]^{2-}$ can be synthesized easily by passing H_2S gas through a strongly basic aqueous solution of molybdate or tungstate ions:

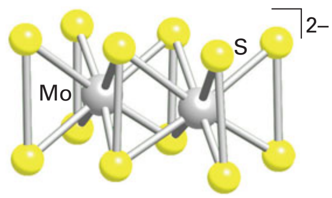
$$[MoO_4]^{2-}(aq) + 4\,H_2S(g) \rightarrow [MoS_4]^{2-}(aq) + 4\,H_2O(l)$$

These tetrathiometallate anions are building blocks for the synthesis of complexes containing more metal atoms. For example, they will coordinate to many dipositive metal ions, such as Co^{2+} and Zn^{2+}:

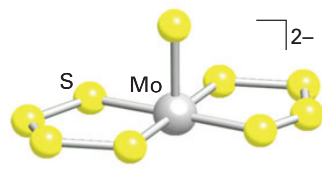
$$Co^{2+}(aq) + 2\,[MoS_4]^{2-}(aq) \rightarrow [S_2MoS_2CoS_2MoS_2]^{2-}(aq)$$

The polysulfide ions, such as S_2^{2-} and S_3^{2-}, which are formed by addition of elemental sulfur to a solution of ammonium sulfide, can also act as ligands. An example is $[Mo_2(S_2)_6]^{2-}$ (**11**), which is formed from ammonium polysulfide and MoO_4^{2-}; it contains side-bonded S_2^{2-} ligands. The larger polysulfides bond to metal atoms forming chelate rings, as in $[MoS(S_4)_2]^{2-}$ (**12**), which contains chelating S_4^{2-} ligands.

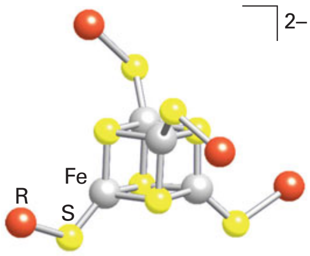
Clusters of Fe and S atoms ('Fe—S clusters') are cofactors in many enzymes, including nitrogenase, which converts N_2 to NH_3 under ambient conditions (Chapter 15). Synthetic analogues such as the cubane shown in (**13**) have been prepared and characterized. The cubane contains Fe and S atoms on alternate corners, so each S atom bridges three Fe



11 $[Mo_2(S_2)_6]^{2-}$



12 $[MoS(S_4)_2]^{2-}$



13 $[Fe_4S_4(SR)_4]^{2-}$

atoms; each of the Fe atoms has a thiolate group, RS^-, occupying a terminal position. These compounds can be obtained from simple starting materials in the absence of air using a polar aprotic solvent such as dimethylsulfoxide:

$$4\,FeCl_3 + 4\,NaSH + 6\,RSH + 10\,NaOCH_3 \rightarrow [Fe_4S_4(SR)_4]^{2-} + RS{-}SR + 14\,Na^+ + 12\,Cl^- + 10\,CH_3OH$$

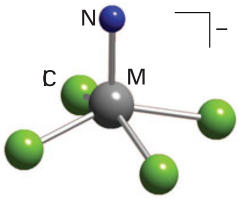
The HS^- ion provides the sulfide ligands for the cage, the RS^- ion serves as both a ligand and a reducing agent, and the CH_3O^- ion acts as a base. Addition of the bulky cation from $(Et_4N)Cl$ results in the formation of black crystals of $(Et_4N)_2[Fe_4S_4(SR)_4]$. The observation that this reaction is successful, in good yield, with many different R groups indicates that the Fe_4S_4 cage is thermodynamically more stable than other possibilities. The cage undergoes reversible one-electron reduction to $[Fe_4S_4(SR)_4]^{3-}$, a reaction that is important in biological oxidation and reduction (Section 27.8).

19.10 Nitrido and alkylidyne complexes

Key points: Multiple M−L bonds are common with high oxidation state metals of the early d-metal series; these bonds weaken the bonds to the trans ligands.

The nitrido ligand (N≡) and alkylidyne ligand (RC≡, previously known as carbido) are present formally as N^{3-} and RC^{3-}. Thus, complexes containing them are usually of very high oxidation state. The highly reactive compound $(Me_3CO)_3W{\equiv}W(OCMe_3)_3$ cleaves the C≡N bond in benzonitrile to give both a nitrido and an alkylidyne ligand:

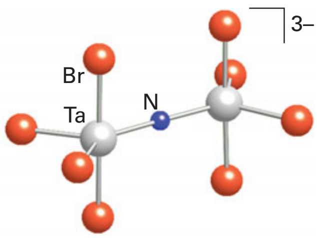
$$(Me_3CO)_3W{\equiv}W(OCMe_3)_3 + PhC{\equiv}N \rightarrow (Me_3CO)_3W{\equiv}CPh + (Me_3CO)_3W{\equiv}N$$



14 $[M(N)Cl_4]^-$

The remarkable feature of this reaction is that the C≡N bond is broken despite its considerable strength (890 kJ mol^{-1}), so the W≡C and W≡N bond enthalpies must be substantial.

Both nitrido and alkylidyne complexes are known to have short M−N and M−C bonds, which confirms the existence of multiple metal−ligand bonds. Like the O^{2-} group, these ligands are strong π donors and weaken the *trans* metal−ligand bond. The order of this influence is RC≡ > N≡ > O=; this weakening is attributed to competition for π donation into a common set of metal d orbitals.



15 $[Br_4Ta(\mu\text{-}N)TaBr_4]^{3-}$

Many nitrido complexes are known for the elements in Groups 5−8. They are most numerous for Mo and W in Group 6, Re in Group 7, and Ru and Os in Group 8. Square-pyramidal nitrido complexes of formula $[MNX_4]^-$ are known for M(VI), with M = Mo, Re, Ru, and Os, and X = F, Cl, Br, and (in some cases) I. These square-pyramidal structures (**14**) have short M≡N bonds (157 to 166 pm). In a few cases, a ligand is attached in the sixth site, but the resulting bond is long and weak, like the oxido complexes discussed earlier. Examples of M=N=M species are known, such as $[Ta_2NBr_8]^{3-}$ (**15**), but nitrogen analogues of the polyoxometallates described earlier are unknown.

19.11 Metal−metal bonded compounds and clusters

The traditional view of the chemical properties of the d metals is from the perspective of ionic solids and complexes that contain a single central metal ion surrounded by a set of ligands. However, with the development of improved techniques for the determination of structure, it has been recognized that there are also many d-metal compounds that have metal−metal (M−M) bond distances comparable to or shorter than those in the elemental metal. These findings have stimulated additional X-ray structural investigations of compounds that might contain M−M bonds and have inspired research into syntheses designed to broaden the range of such **cluster compounds**. Examples of these metal (and nonmetal) cluster compounds are now known in every block of the periodic table (Fig. 19.16), but they are most numerous in the d block.

A rigorous definition of metal clusters restricts them to molecular complexes with metal−metal bonds that form triangular or larger structures. This definition, however, would exclude linear M−M compounds, and is normally relaxed. We shall consider any M−M bonded system to be a cluster.

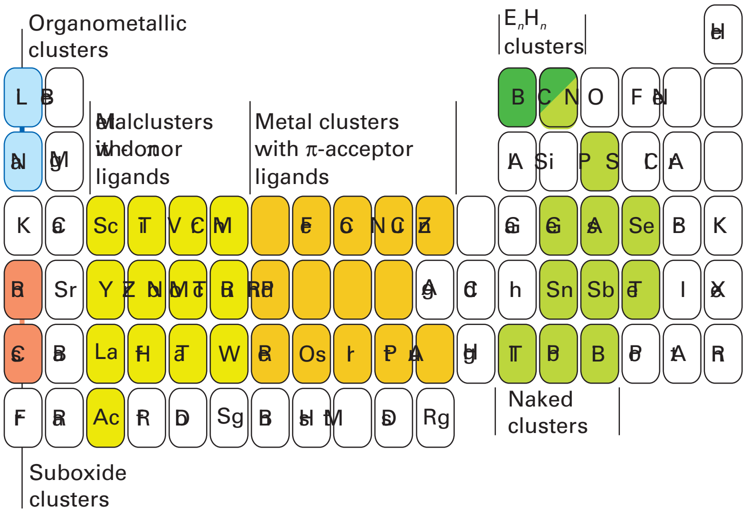


Figure 19.16 The major classes of cluster-forming elements. Note that carbon is in two classes (for example it occurs as C_8H_8 and as C_{60}). (Adapted from D.M.P. Mingos and D.J. Wales, *Introduction to cluster chemistry*. Prentice Hall, Englewood Cliffs (1990).)

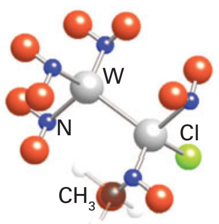
(a) Metal–metal bonds

Key point: Metal–metal bonds with bond orders up to five are formed by many d metals in low oxidation states.

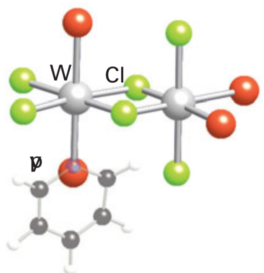
The first d-block metal–metal bonded species to be identified was the Hg_2^{2+} ion of mercury(I) compounds, as occurs in Hg_2Cl_2, and examples of metal–metal bonded compounds and clusters are now known for most of the d metals. Some of their common structural motifs are an ethane-like structure (**16**), an edge-shared bioctahedron (**17**), a face-shared bioctahedron (**18**), and the tetragonal prism of $[Re_2Cl_8]^{2-}$ (**19**).

If we consider the possible overlap between d orbitals on adjacent metal atoms, we can see that (Fig. 19.17):

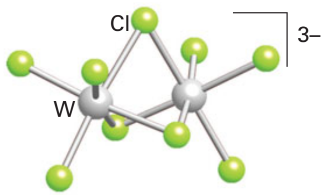
- a σ bond between two metal atoms can arise from the overlap of a d_{z^2} orbital from each atom
- two π bonds can arise from the overlap of d_{zx} or d_{yz} orbitals
- two δ bonds can be formed from the overlap of two face-to-face d_{xy} or $d_{x^2-y^2}$ orbitals.



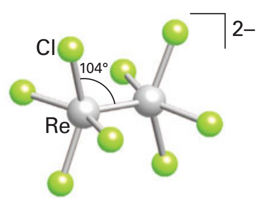
16 $[(Me_2N)_3WWCl(NMe_2)_2]$, Me = CH_3



17 $[(Cl)_2(py)_2W(\mu\text{-}Cl)_2W(Cl)_2(py)_2]$



18 $[(Cl)_3W(\mu\text{-}Cl)_3W(Cl)_3]^{3-}$



19 $[Re_2Cl_8]^{2-}$

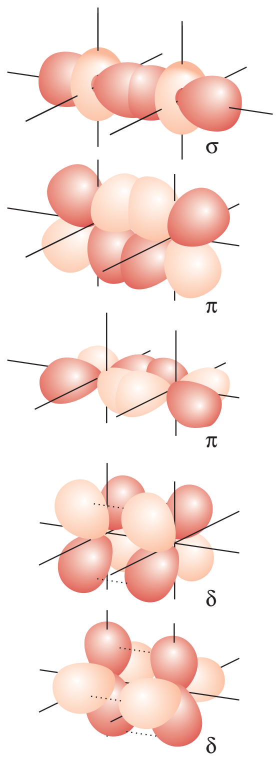


Figure 19.17 The origin of σ, π, and δ interactions between the d orbitals of two d-metal atoms situated along the *z*-axis. Only bonding combinations are shown.

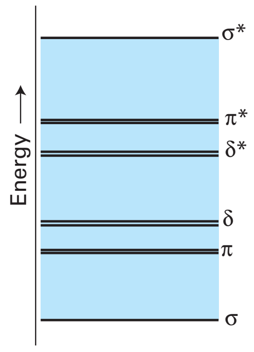
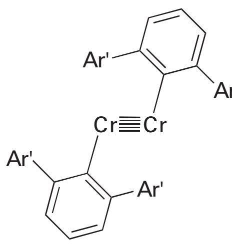


Figure 19.18 Approximate molecular orbital energy level scheme for M—M interactions.



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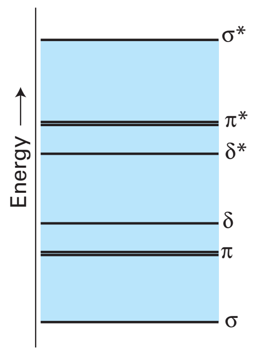
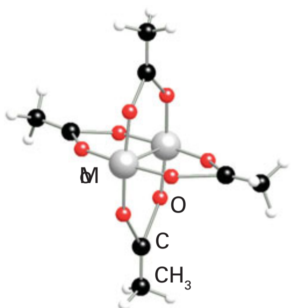


Figure 19.19 Approximate molecular orbital energy level scheme for the M—M interactions in a quadruply bonded system, where only the $d_{x^2-y^2}$ is utilized in bonding to the ligands.



21 $[Mo_2(\mu\text{-}CH_3CO_2)_4]$

Thus a quintuple bond could result if all the bonding orbitals are occupied to give the electron configuration $\sigma^2\pi^4\delta^4$ (Fig.19.18).

Five d electrons would be needed from each metal for a quintuple bond, and this is exactly the case for the d^5 Cr(I) centres in (**20**). In this molecule, the two Cr atoms are separated by a very short distance of 183.5 pm, unsupported by any additional bridging ligand interactions; compare this distance with the separation of Cr atoms in the bulk metal, which is 258 pm. The Re compound (**19**) also lacks bridging ligand interactions, but here the two d^4 Re(III) centres can form only a quadruple Re—Re bond. The four d electrons from each Re atom in (**19**) result in the configuration $\sigma^2\pi^4\delta^2$, and it is believed that the $d_{x^2-y^2}$ orbital is involved in bonding to the Cl^- ligands. Evidence for quadruple bonding comes from the observation that $[Re_2Cl_8]^{2-}$ has an eclipsed array of Cl ligands, which is sterically unfavourable. It is argued that the δ bond, which is formed only when the d_{xy} orbitals are confacial, locks the complex in the eclipsed conformation.

Many other species with multiple metal—metal bonds, where the $d_{x^2-y^2}$ orbital is involved in bonding to ligand species, are known. In all these complexes, the molecular orbital diagram shown in Figure 19.19 becomes appropriate and the maximum possible bond order is 4. A well-known example is the quadruply bonded compound molybdenum(II) acetate (**21**), which is prepared by heating $Mo(CO)_6$ with acetic acid:

$$2\,Mo(CO)_6 + 4\,CH_3COOH \rightarrow Mo_2(O_2CCH_3)_4 + 2\,H_2 + 12\,CO$$

The dimolybdenum complex is an excellent starting material for the preparation of other Mo—Mo compounds. For example, the quadruply bonded chlorido complex is obtained when the acetato complex is treated with concentrated hydrochloric acid at below room temperature:

$$Mo_2(O_2CCH_3)_4(aq) + 4\,H^+(aq) + 8\,Cl^-(aq) \rightarrow [Mo_2Cl_8]^{4-}(aq) + 4\,CH_3COOH(aq)$$

As shown in Table 19.8, incomplete occupation of the bonding orbitals can result in a reduction of the formal bond order to 3.5 or to the triply bonded M≡M systems. These complexes are more numerous than the quadruply bonded complexes and, because δ bonds are weak, M≡M bond lengths are often similar to those of quadruply bonded systems. A decrease of bond order can also stem from the occupation of both the δ* orbitals and, once these are fully occupied, successive occupation of the two higher lying π* orbitals leads to further decrease in the bond order from 2.5 to 1.

As with carbon—carbon multiple bonds, metal—metal multiple bonds are centres of reaction. However, the variety of structures resulting from the reactions of metal—metal multiple bonded compounds is more diverse than for organic compounds. For example:

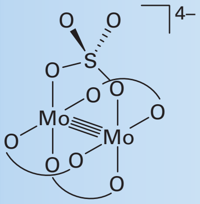
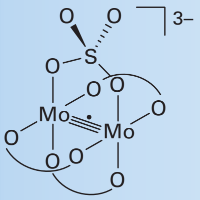
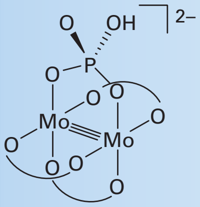
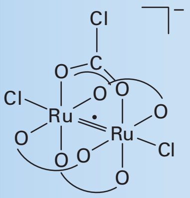
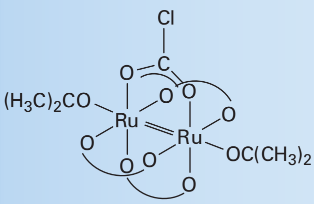
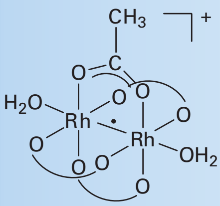
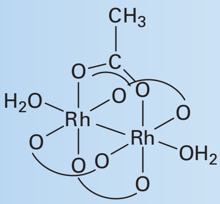
In this reaction, HI adds across a triple bond but both the H and I bridge the metal atoms; the outcome is quite unlike the addition of HX to an alkyne, which results in a substituted alkene. The reaction product can be regarded as containing a 3*c*,2*e* MHM bridge and an iodide anion bonding by two conventional 2*c*,2*e* bonds, one to each Mo atom.

Larger metal clusters can be synthesized by addition to a metal—metal multiple bond. For example, $Pt(PPh_3)_4$ loses two triphenylphosphine ligands when it adds to the Mo≡Mo triple bond, resulting in a three-metal cluster:

(b) Clusters

Key points: **Metal clusters may be formed; for early d-block elements they are favoured by good donor ligands, whereas for later d-block elements π-acceptor ligands are needed.**

Table 19.8 Examples of metal—metal bonded tetragonal prismatic complexes*

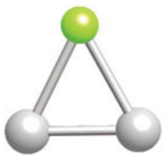
Complex	Configuration	Bond order	M—M bond length/pm
	$\sigma^2\pi^4\delta^2$	4	211
	$\sigma^2\pi^4\delta^1$	3.5	217
	$\sigma^2\pi^4$	3	222
	$\sigma^2\pi^4\delta^2\delta^{*1}\pi^{*2}$	2.5	227
	$\sigma^2\pi^4\delta^2\delta^{*2}\pi^{*2}$	2	238
	$\sigma^2\pi^4\delta^2\delta^{*1}\pi^{*4}$	1.5	232
	$\sigma^2\pi^4\delta^2\delta^{*2}\pi^{*4}$	1	239

* When multiple bridging ligands are present, only one is shown in detail.

Figure 19.16 showed how cluster compounds may be classified according to ligand type. Although this classification is not appropriate in detail for all cluster compounds, it provides information on the principal ligand types and some insight into the bonding. For example, alkyllithium compounds often have alkyl groups attached to a metal cluster by two-electron multicentre bonds. Clusters of the early d-block elements and the lanthanoids generally contain donor ligands such as Br^-. These ligands can fill some of the low-lying orbitals of these electron-poor metal atoms by σ- and π-electron donation. In contrast, electron-rich metal clusters on the right of the d block generally contain π-acceptor ligands such as CO, which remove some of the electron density from the metal. Many p-block elements do not require ligands to complete the valence shells of the individual atoms in clusters and can exist as **naked clusters**, which are clusters of the form E_n with no accompanying ligands. The ionic clusters Pb_5^{2-} and Sn_9^{4-} are two examples of naked clusters formed by metals.

Clusters may involve M–M bonds within a discrete molecular cluster or in extended solid-state compounds. When no bridging ligands are present in a cluster, as in $[Re_2Cl_8]^{2-}$ (**19**), the presence of a metal–metal bond is unambiguous. When bridging ligands are present, careful observations and measurement (typically of bond lengths and magnetic properties) are needed to identify direct metal–metal bonding. We shall see in Section 22.20 that there is an extensive range of organometallic metal cluster compounds of the middle-to-late d-block elements that are stabilized by π-acceptor ligands, particularly CO.


22


23


24

The bonding patterns in most metal cluster compounds are so intricate that metal–metal bond strengths cannot be determined with great precision. Some evidence, however, such as the stability of compounds and the magnitudes of M–M force constants, indicates that there is an increase in M–M bond strength down a group, perhaps on account of the greater spatial extension of d orbitals in heavier atoms. This trend may be the reason why there are so many more metal–metal bonded compounds for the 4d- and 5d-series metals than for their 3d-series counterparts. Figure 19.1 indicates that for the bulk metals, the metal–metal bonds in the d block are strongest in the 4d and 5d series, and this feature carries over into their compounds. By contrast, element–element bonds weaken down a group in the p block.

Not all the early d-block metal–metal bonded compounds are discrete clusters. Many extended metal–metal bonded compounds exist, such as the multiple-chain scandium subhalides, Sc_7Cl_{10} (Fig. 19.20) and Sc_5Cl_6, and layered compounds such as ZrCl (Fig. 19.21). In ZrCl, the Zr atoms are within bonding distance in two adjacent layers of metal atoms sandwiched between Cl^- layers. We have already seen that Sc and Zr adopt their group oxidation states (+3 and +4, respectively) when exposed to air and moisture, so these metal–metal bonded compounds are prepared out of contact with air and moisture. For example, ZrCl is prepared by reducing zirconium tetrachloride with zirconium metal in a sealed tantalum tube at high temperatures:

$$3Zr(s) + ZrCl_4(g) \xrightarrow{600-800^\circ C} 4ZrCl(s)$$

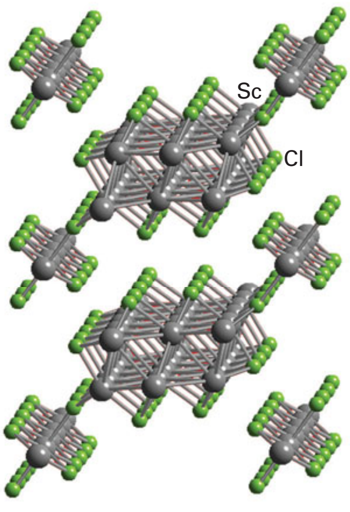


Figure 19.20 The structure of Sc_7Cl_{10}.

Discrete clusters—as distinct from the extended M–M bonded solid-state compounds just described—are often soluble, and can be manipulated in solution. The π-donor ligands in these clusters typically occur in one of several locations, namely, in a terminal position (**22**), bridging two metal atoms (**23**), or bridging three metal atoms (**24**). Clusters may also be linked by bridging halides or chalcogenides in the solid state. For example, the solid compound of formula $MoCl_2$ consists of octahedral Mo clusters linked by Cl bridges.

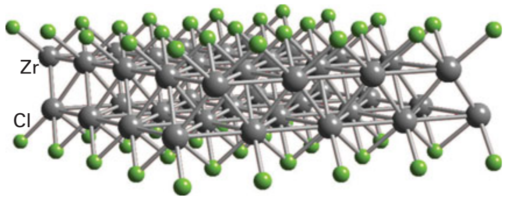


Figure 19.21 The structure of ZrCl consists of layers of metal atoms in graphite-like hexagonal nets.

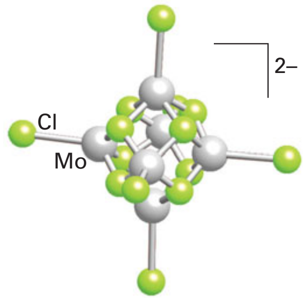
The bridged structure of $MoCl_2$ is in sharp contrast to $CrCl_2$, which has discrete Cr atoms arranged in a distorted rutile structure. Samples of $MoCl_2$ may be prepared in a sealed glass tube by the reaction of $MoCl_5$ in a mixture of molten $NaAlCl_4$ and $AlCl_3$, together with aluminium metal as the reducing agent:

$$MoCl_5(s) + Al(s) \xrightarrow{NaAlCl_4/AlCl_3(l), 200°C} MoCl_2(s) + AlCl_3(l)$$

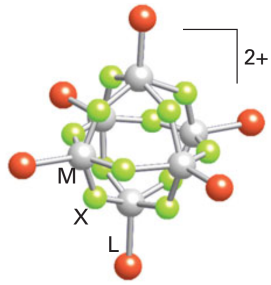
The compound can withstand oxidation under mild conditions. When treated with hydrochloric acid it forms the anionic cluster $[Mo_6Cl_{14}]^{2-}$. This cluster contains an octahedral array of Mo atoms with a Cl atom bridging each triangular face and a terminal Cl atom on each Mo vertex (**25**). These terminal Cl atoms can be replaced by other halogens, alkoxides, and phosphines. An analogous series of tungsten cluster compounds is known. Once formed, these Mo and W compounds can be handled in air and water at room temperature on account of their kinetic barriers to decomposition. The oxidation number of the metal is +2, so the metal valence electron count is $(6 - 2) \times 6 = 24$. One-electron oxidation and reduction of the clusters is possible.

Similar octahedral clusters are known for Nb and Ta in Group 5, and for Zr in Group 4. The cluster $[Nb_6Cl_{12}L_6]^{2+}$ and its Ta analogue have an octahedral framework with edge-bridging Cl atoms and six terminal ligands (**26**). The metal valence electron count in this case is 16, but these clusters can be oxidized in several steps. One example from Group 4 is $[Zr_6Cl_{18}C]^{4-}$, which has the same metal and Cl atom array together with a C atom in the centre of the octahedron.

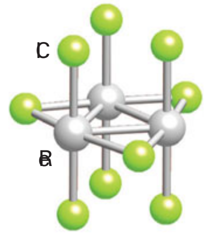
Rhenium trichloride, which consists of Re_3Cl_9 clusters linked by weak halide bridges (**27**), provides the starting material for the preparation of a series of M_3 clusters that have been studied thoroughly. As with molybdenum dichloride, the intercluster bridges in the solid state can be broken by reaction with potential ligands. For example, treatment of Re_3Cl_9 with Cl^- ions produces the discrete complex $[Re_3Cl_{12}]^{3-}$. Neutral ligands, such as trialkylphosphines, can also occupy these coordination sites and result in clusters of the general formula $Re_3Cl_9L_3$.



25 $[Mo_6Cl_{14}]^{2-}$



26 $[M_6X_{12}L_6]^{2+}$



27 Re_3Cl_9

EXAMPLE 19.3 Metal–metal bonding and clusters

Suggest which interactions might be responsible for, and thus the bond order of, the metal–metal bond in the Hg_2^{2+} ion.

Answer We need to judge the types of bonds that can form from the available atomic orbitals on each metal atom and the bond order that results from their occupation. The oxidation state of mercury in Hg_2^{2+} is Hg(I) and therefore its electron configuration is $d^{10}s^1$. Although overlap of the d orbitals in the manner depicted in Fig. 19.17 is possible, the 20 d electrons from the two Hg ions would result in complete filling of both the bonding and the antibonding orbitals, with no effective bonding. Therefore the bonding must come from the overlap of s orbitals on each ion and the two remaining s electrons: σ bonding and antibonding orbitals can be constructed and as only the former is occupied, it results in a single Hg–Hg bond. Even if a degree of sd hybridization is invoked, the description still demands the involvement of s orbitals in the bonding and the bond order remains 1.

Self-test 19.3 Describe the probable structure of the compound formed when Re_3Cl_9 is dissolved in a solvent containing PPh_3.

FURTHER READING

D.M.P. Mingos, *Essential trends in inorganic chemistry*. Oxford University Press (1998). A survey of inorganic chemistry from the perspective of structure and bonding.

R.B. King (ed.), *Encyclopedia of inorganic chemistry*. Wiley (2005).

M.T. Pope, Polyoxoanions: synthesis and structure. In *Comprehensive coordination chemistry II*, Vol. 4 (ed. J.A. McCleverty and T.J. Meyer), Chapter 10. Elsevier (2004). A discussion of metal oxides in solution, focusing on their tendency to form polyoxoanions.

M.H. Chisholm (ed.) *Early transition metal clusters with π-donor ligands*. VCH, Weinheim (1995).

EXERCISES

19.1 Without reference to a periodic table, sketch the first series of the d block, including the symbols of the elements. Indicate those elements for which the group oxidation number is common by C, those for which the group oxidation number can be reached but is a powerful oxidizing agent by O, and those for which the group oxidation number is not achieved by N.

19.2 Explain why the enthalpy of sublimation of Re(s) is significantly greater than that of Mn(s).

19.3 State the trend in the stability of the group oxidation state on descending a group of metallic elements in the d block. Illustrate the trend using standard potentials in acidic solution for Groups 5 and 6.

19.4 For each part, give balanced chemical equations or NR (for no reaction) and rationalize your answer in terms of trends in oxidation states.

(a) $Cr^{2+}(aq) + Fe^{3+}(aq) \rightarrow$

(b) $CrO_4^{2-}(aq) + MoO_2(s) \rightarrow$

(c) $MnO_4^-(aq) + Cr^{3+}(aq) \rightarrow$

19.5 (a) Which ion, $Ni^{2+}(aq)$ or $Mn^{2+}(aq)$, is more likely to form a sulfide in the presence of H_2S? (b) Rationalize your answer with the trends in hard and soft character across Period 4. (c) Give a balanced chemical equation for the reaction.

19.6 Preferably without reference to the text (a) write out the d block of the periodic table, (b) indicate the metals that form difluorides with the rutile or fluorite structures, and (c) indicate the region of the periodic table in which metal–metal bonded halide compounds are formed, giving one example.

19.7 Write a balanced chemical equation for the reaction that occurs when *cis*-$[RuLCl(OH_2)]^+$ (see Fig. 19.9) in acidic solution at +0.2 V is made strongly basic at the same potential. Write a balanced equation for each of the successive reactions when this same complex at pH = 6 and +0.2 V is exposed to progressively more oxidizing environments up to +1.0 V. Give other examples and a reason for the redox state of the metal centre affecting the extent of protonation of coordinated oxygen.

19.8 Give plausible balanced chemical reactions (or NR for no reaction) for the following combinations, and state the basis for your answer: (a) $MnO_4^-(aq)$ plus $Fe^{2+}(aq)$ in acidic solution, (b) the preparation of $[Mo_6O_{19}]^{2-}(aq)$ from $K_2MoO_4(s)$, (c) $ReCl_5(s)$ plus $KMnO_4(aq)$, (d) $MoCl_2(s)$ plus warm HBr(aq), (e) TiO(s) with HCl(aq) under an inert atmosphere, (f) Cd(s) added to $Hg^{2+}(aq)$.

19.9 Speculate on the structures of the following species and present bonding models to justify your answers: (a) $[Re(O)_2(py)_4]^+$, (b) $[V(O)_2(ox)_2]^{3-}$, (c) $[Mo(O)_2(CN)_4]^{4-}$, (d) $[VOCl_4]^{2-}$.

19.10 Which of the following are likely to have structures that are typical of (a) predominantly ionic, (b) significantly covalent, (c) metal–metal bonded compounds: NiI_2, $NbCl_4$, FeF_2, PtS, and WCl_2? Rationalize the differences and speculate on the structures.

19.11 Indicate the probable occupancy of σ, π, and δ bonding and antibonding orbitals, and the bond order for the following tetragonal prismatic complexes: (a) $[Mo_2(O_2CCH_3)_4]$, (b) $[Cr_2(O_2CC_2H_5)_4]$, (c) $[Cu_2(O_2CCH_3)_4]$.

19.12 Explain the differences in the following redox couples, measured at 25°C:

MnO_4^-/MnO_2	+1.69 V
TcO_4^-/TcO_2	+0.74 V
ReO_4^-/ReO_2	+0.51 V

19.13 Addition of sodium ethanoate to aqueous solutions of Cr(II) gives a red diamagnetic product. Draw the structure of the product, noting any features of interest.

19.14 Consider the two ruthenium complexes in Table 19.8. Using the bonding scheme depicted in Figure 19.19, confirm the bonding orders and electron configurations given in the table.

PROBLEMS

19.1 An amateur chemist claimed the existence of a new metallic element, grubium (Gr), which has the following characteristics. Metallic Gr reacts with 1 M $H^+(aq)$ in the absence of air to produce $Gr^{3+}(aq)$ and $H_2(g)$. In the absence of air $GrCl_2(s)$ dissolves in 1 M $H^+(aq)$ and very slowly yields $H_2(g)$ plus $Gr^{3+}(aq)$. When $Gr^{3+}(aq)$ is exposed to air $GrO^{2+}(aq)$ is produced. From this information, estimate the range of potentials for (a) the reduction of $Gr^{3+}(aq)$ to Gr(s), (b) the reduction of $Gr^{3+}(aq)$ to $GrCl_2(s)$, and (c) the reduction of $GrO^{2+}(aq)$ to $Gr^{3+}(aq)$. Suggest a known element that fits the description of grubium.

19.2 Compared with the p block, the variation in the properties of the elements of the d block across each period is rather modest. Provide evidence for this statement by reference to isostructural compounds. Speculate on reasons why this statement is true.

19.3 Think about the mechanism of electrical conduction and then suggest how electrical conductivity might vary across the d block.

19.4 Many metal salts can be vaporized to a small extent at high temperatures and their structures studied in the vapour phase by electron diffraction. Speculate on the structures that you might expect to find for the following gas-phase species and present your reasoning: (a) TaF_5, (b) MoF_6.

19.5 Discuss ways in which the triamidoamine ligands $[(RNCH_2CH_2)_3N]^{3-}$ (where R is a bulky substituent) can be used to stabilize nitrido, phosphido, and arsenido groups (see R.R. Schrock, *Acc. Chem. Res.*, 1997, **30**, 9).

19.6 Discuss how the presence of a quintuple bond in (**20**) was confirmed, and how a sextuple bond between two metal atoms might arise (see T. Nguyen, A.D. Sutton, M. Brynda, J.C. Fettinger, G.J. Long, and P.P. Power, *Science* 2005, **310**, 844).